

BFS Traversal

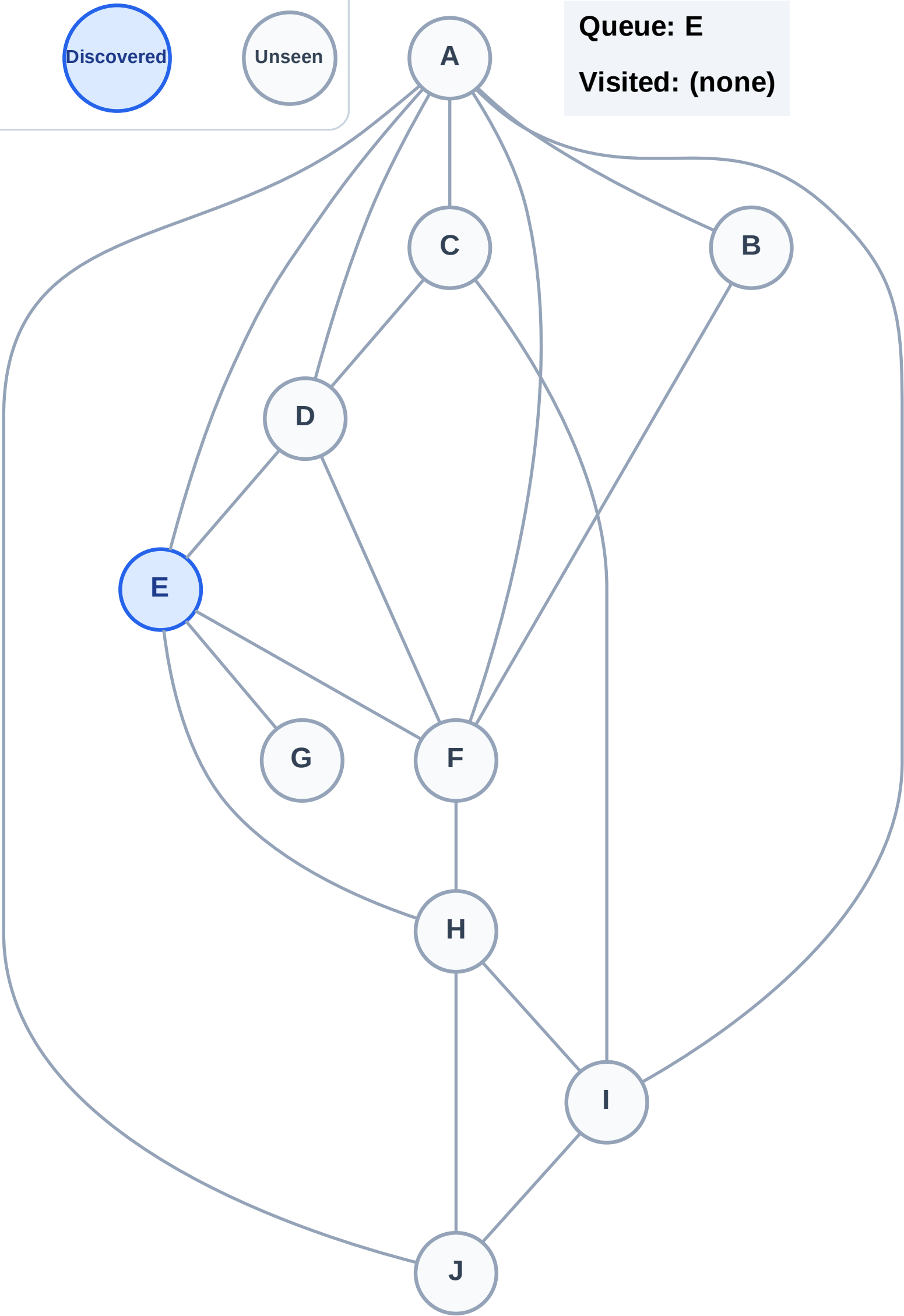
Step 1: Start at E

Legend



Queue: E

Visited: (none)



BFS Traversal

Step 2: Process node E

Legend

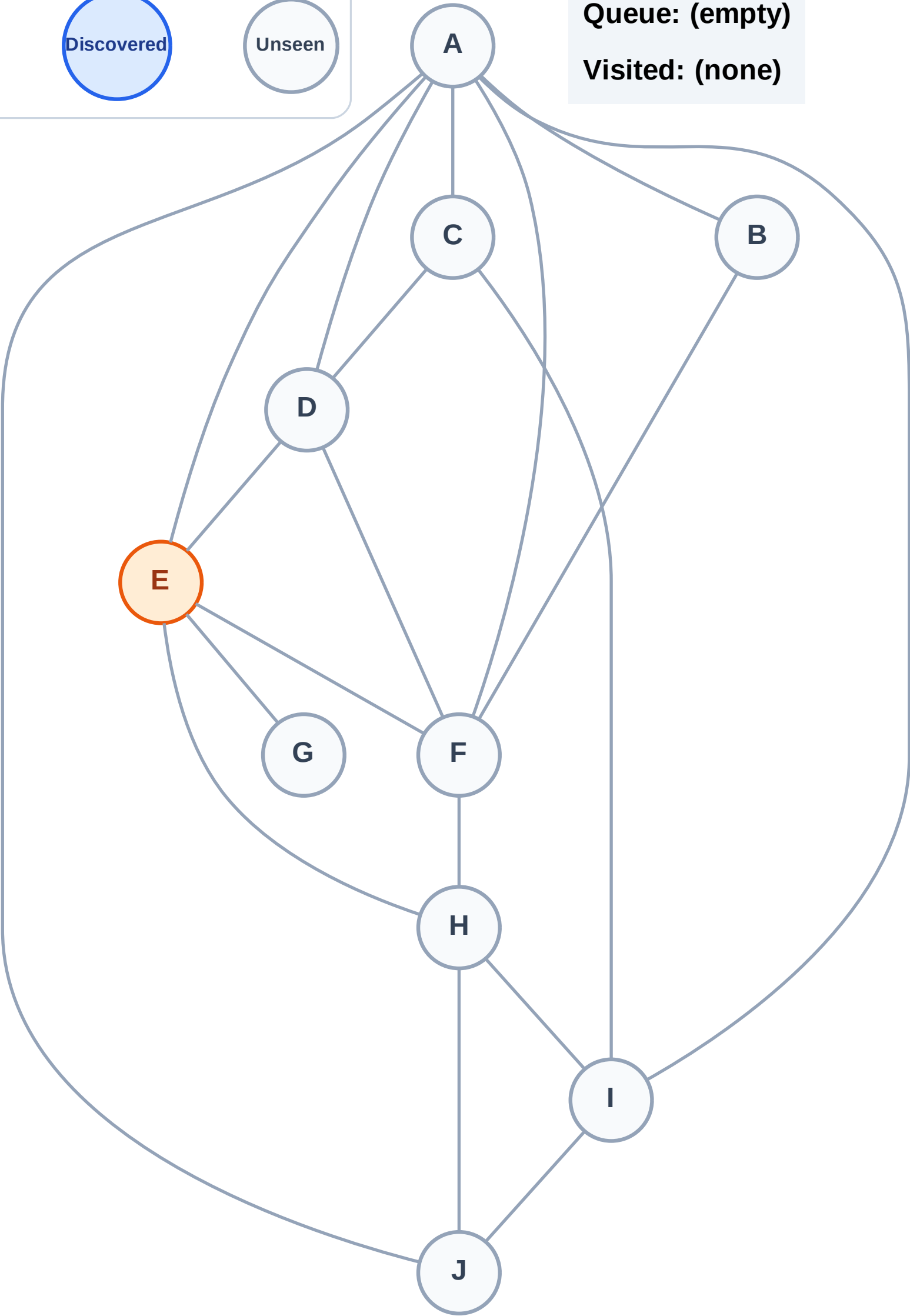
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

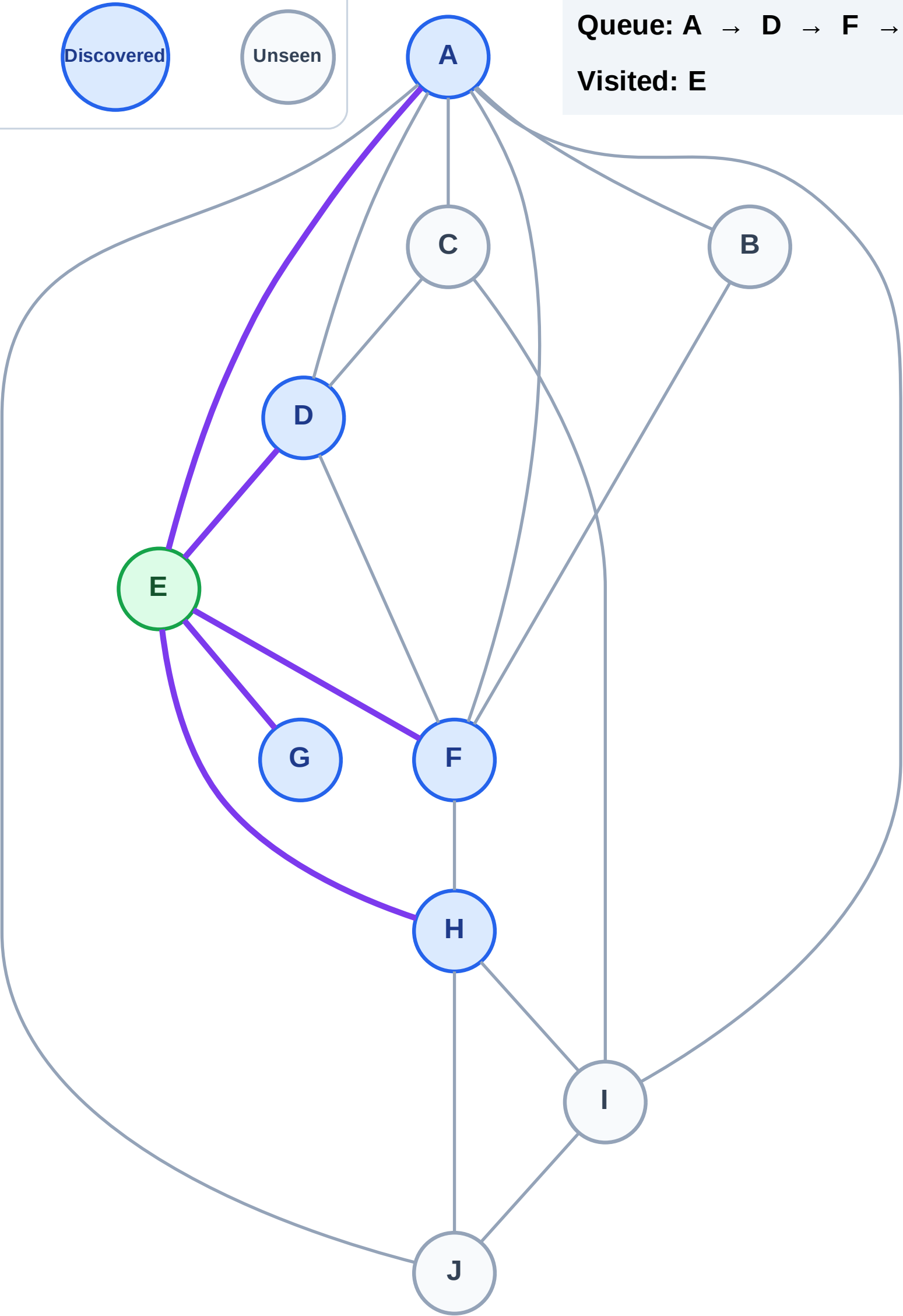
Step 3: Discover A, D, F, G, H from E

Legend



Queue: A → D → F → G → H

Visited: E



BFS Traversal

Step 4: Process node A

Legend

Complete

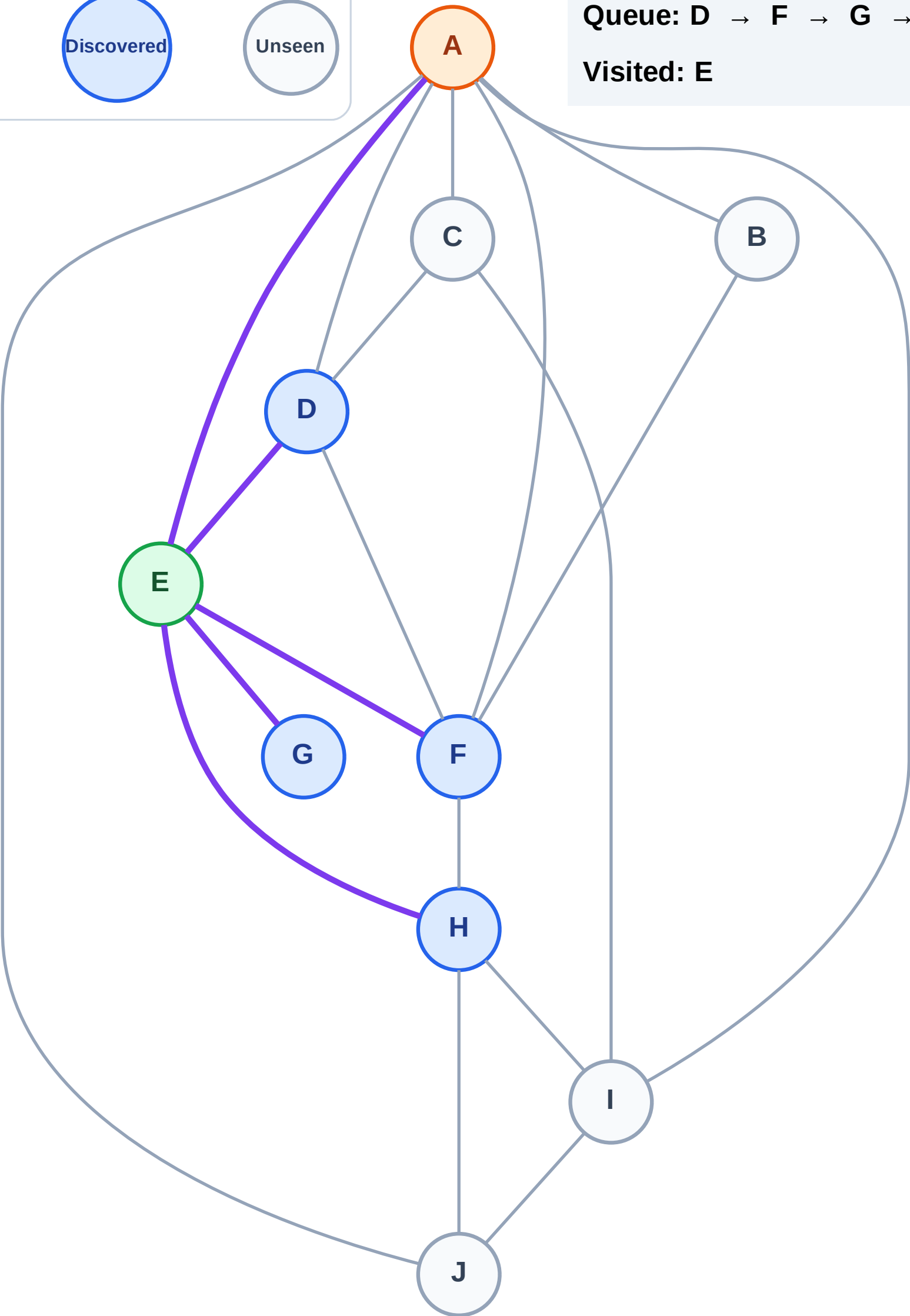
Processing

Discovered

Unseen

Queue: D → F → G → H

Visited: E



BFS Traversal

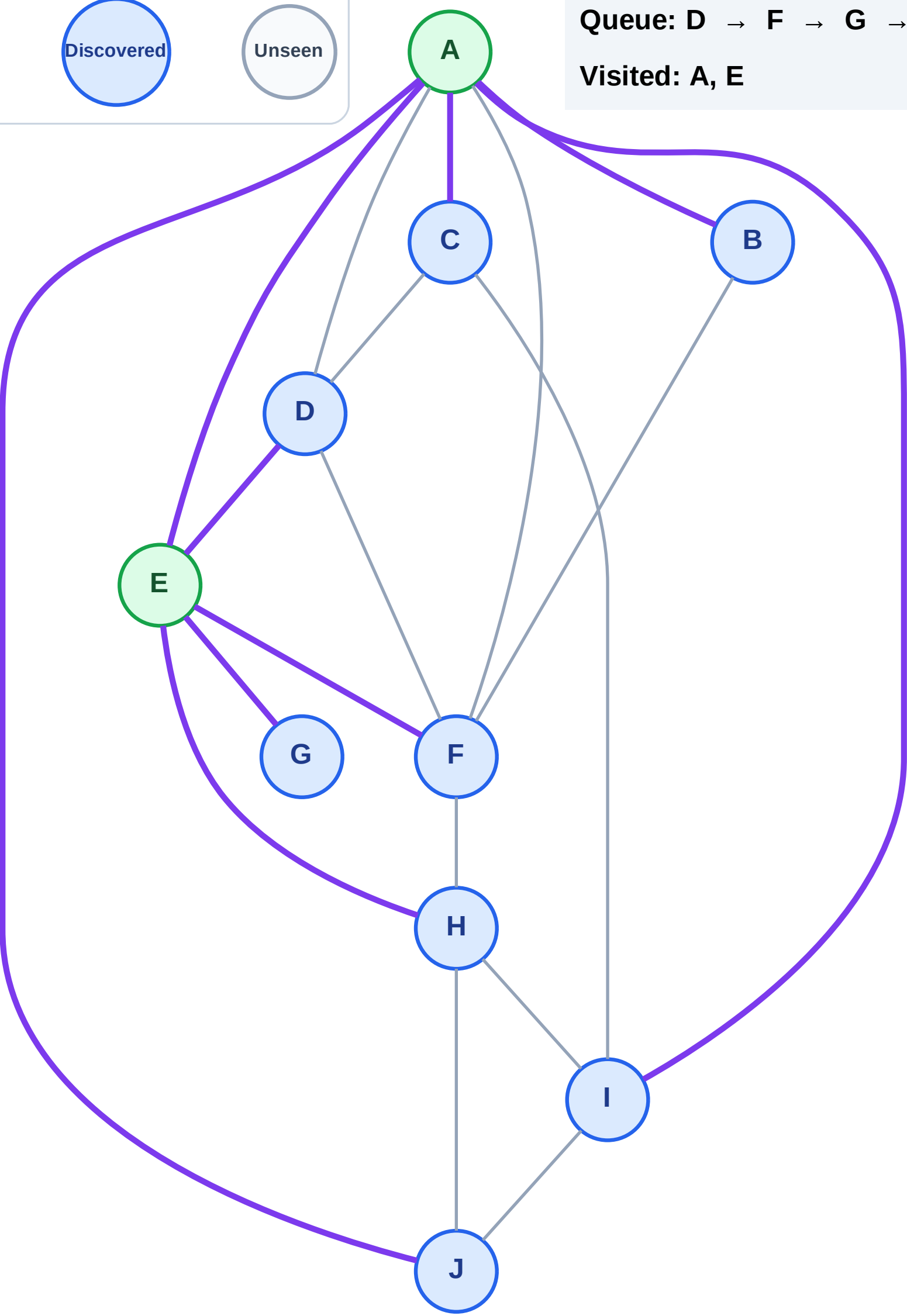
Step 5: Discover B, C, I, J from A

Legend



Queue: D → F → G → H → B → C → I → J

Visited: A, E



BFS Traversal

Step 6: Process node D

Legend

Complete

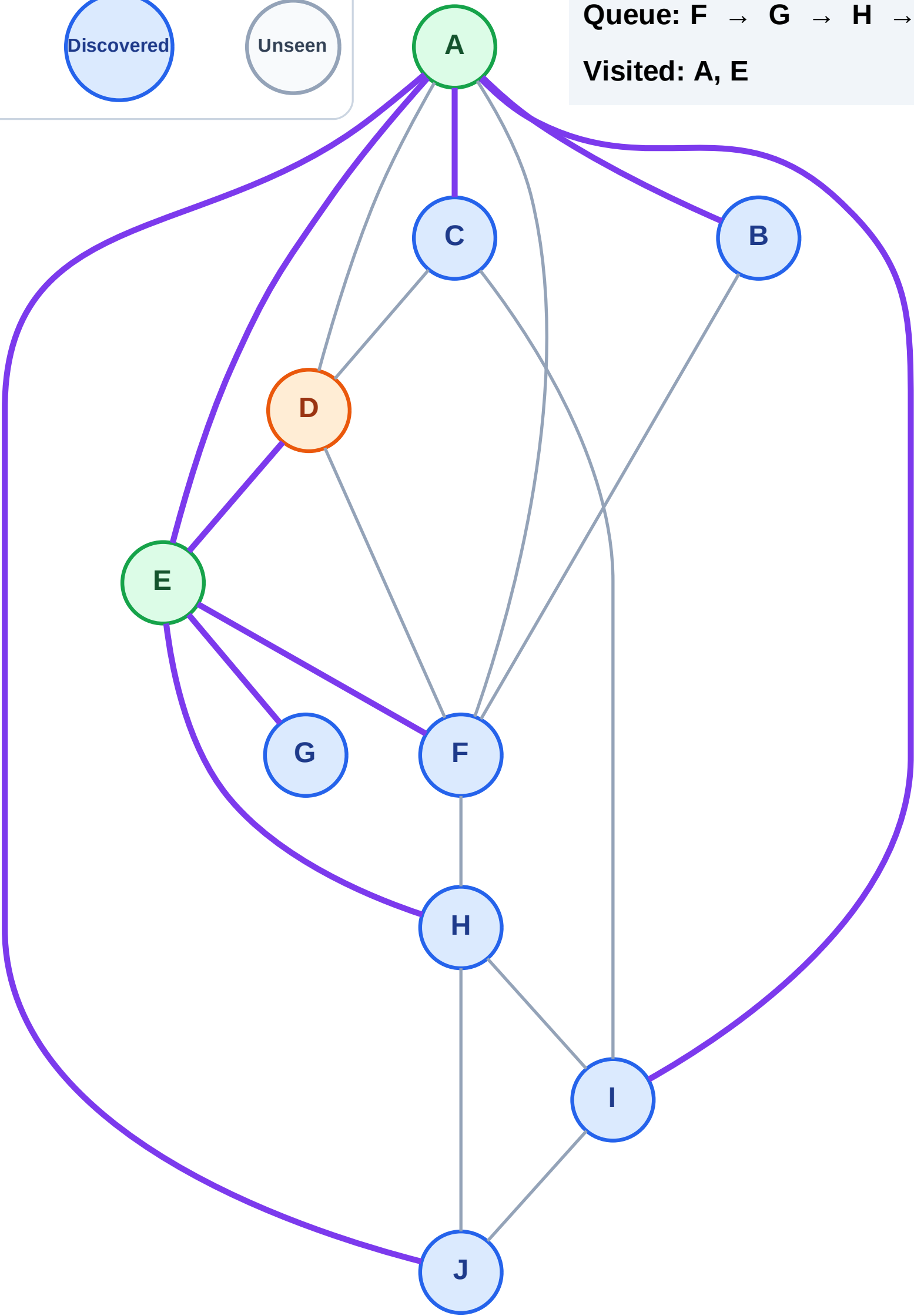
Processing

Discovered

Unseen

Queue: F → G → H → B → C → I → J

Visited: A, E



BFS Traversal

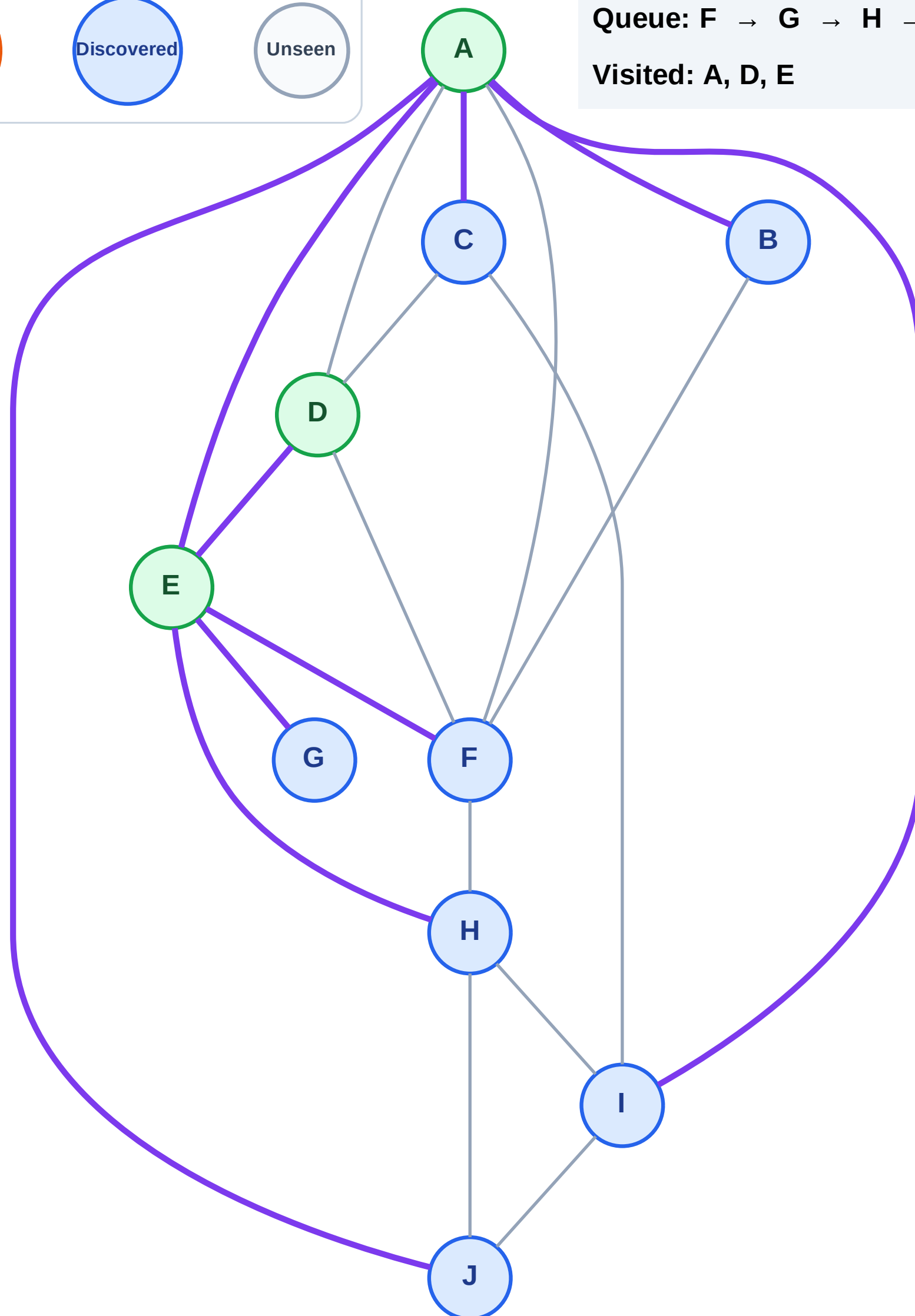
Step 7: No new neighbors from D

Legend



Queue: F → G → H → B → C → I → J

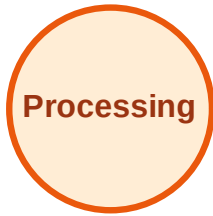
Visited: A, D, E



BFS Traversal

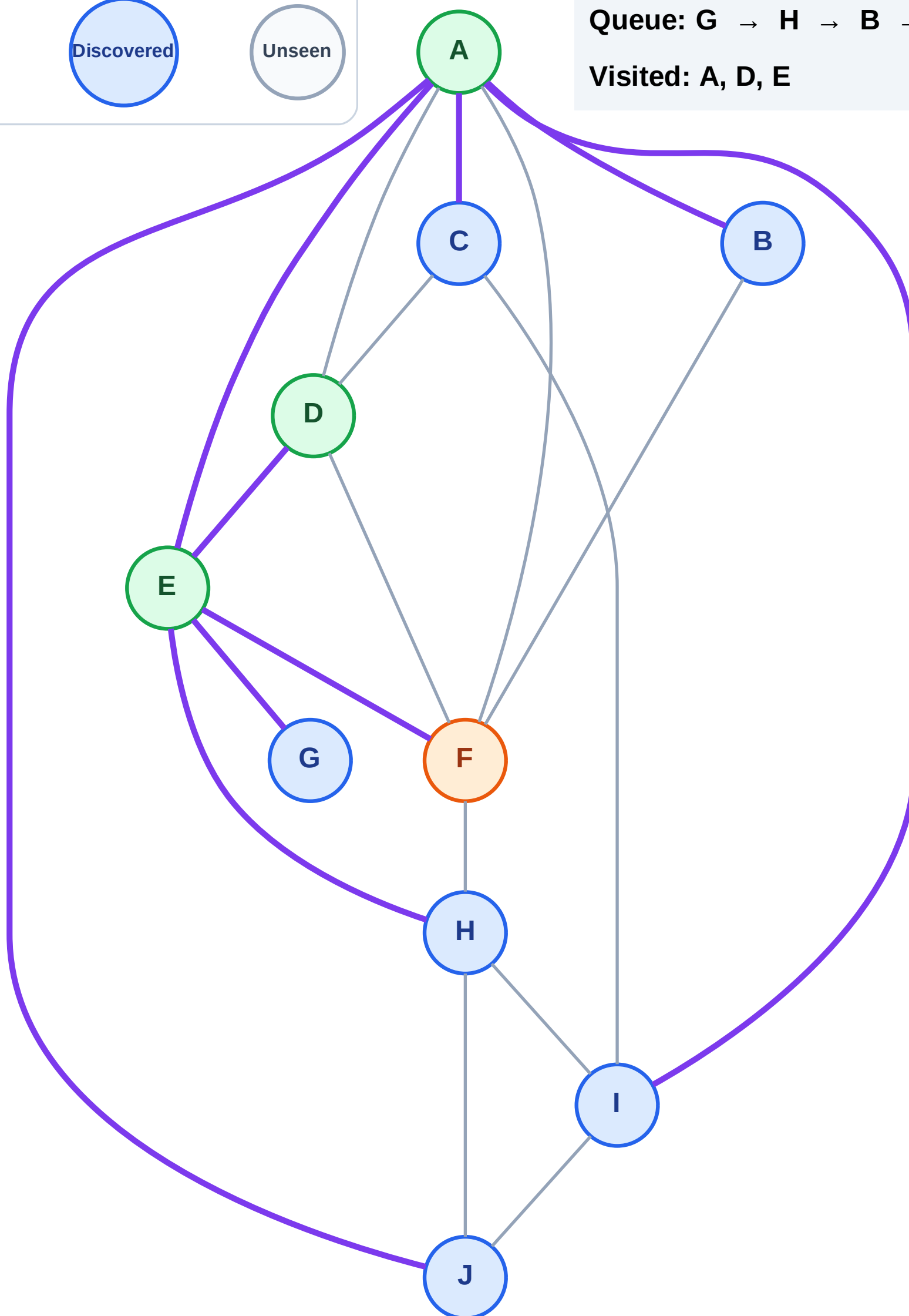
Step 8: Process node F

Legend



Queue: G → H → B → C → I → J

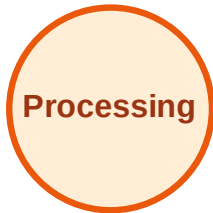
Visited: A, D, E



BFS Traversal

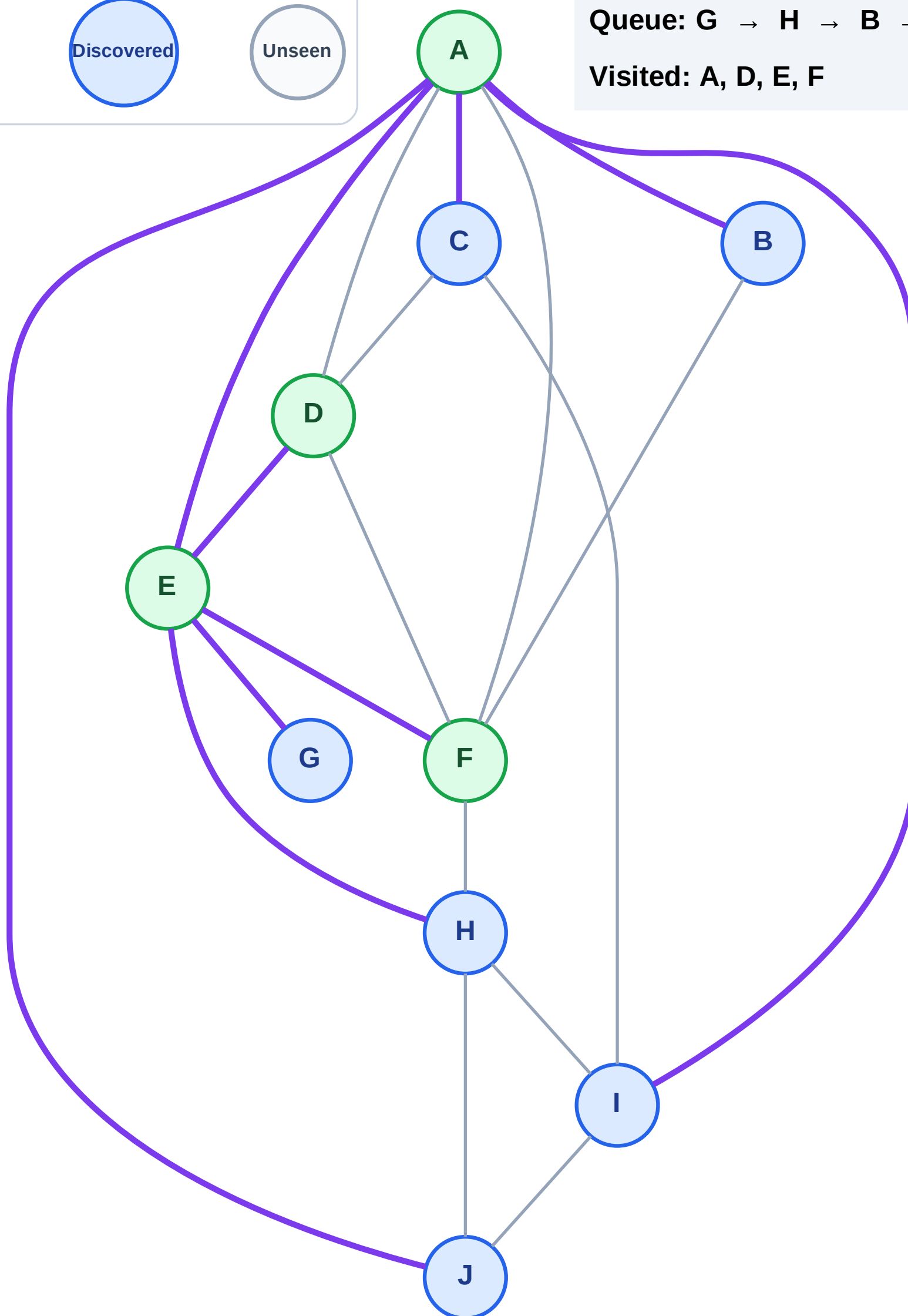
Step 9: No new neighbors from F

Legend



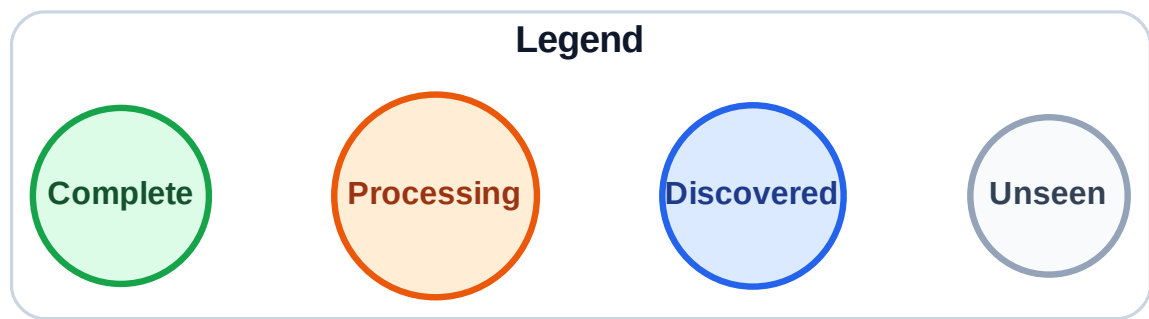
Queue: G → H → B → C → I → J

Visited: A, D, E, F



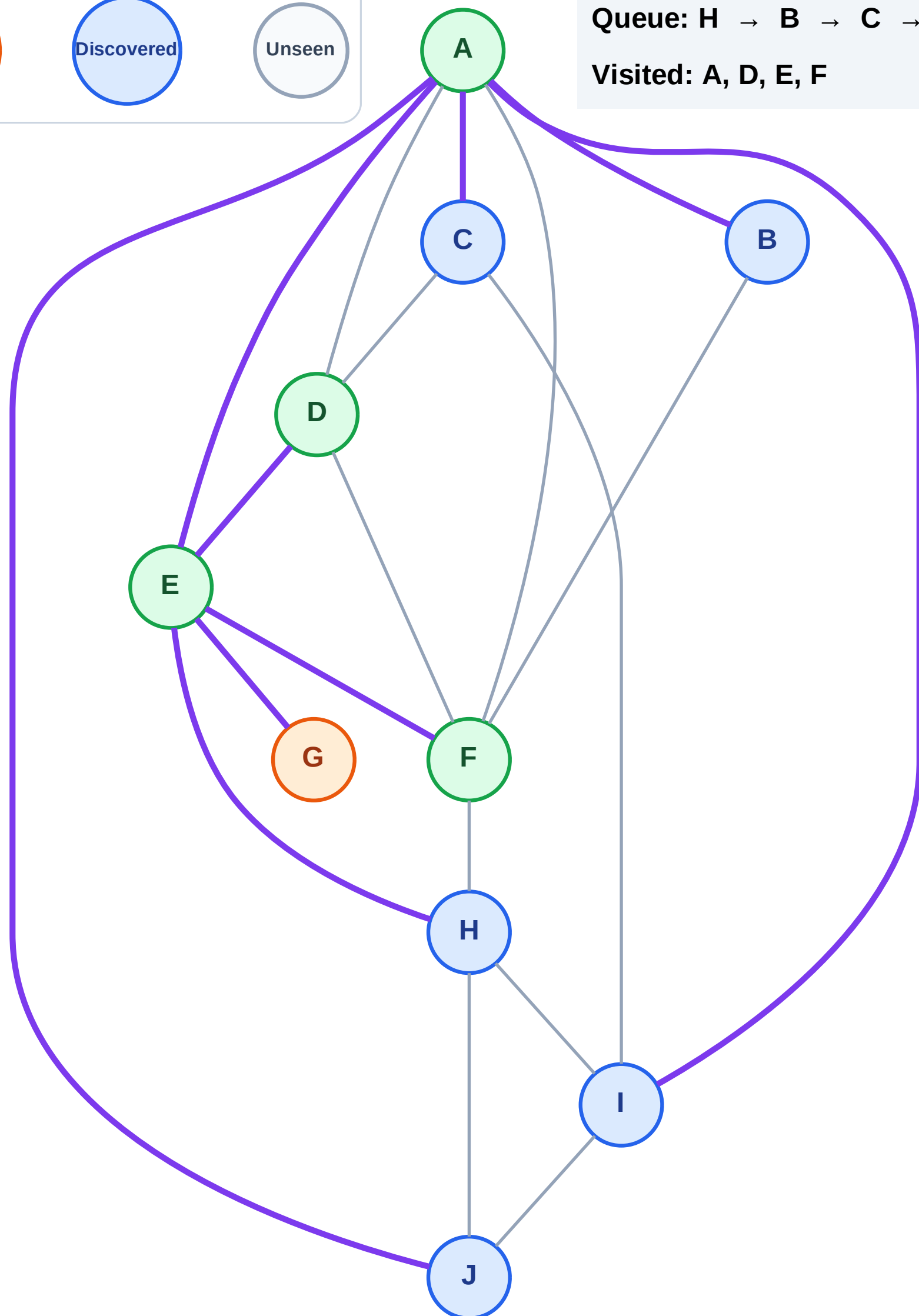
BFS Traversal

Step 10: Process node G



Queue: H → B → C → I → J

Visited: A, D, E, F



BFS Traversal

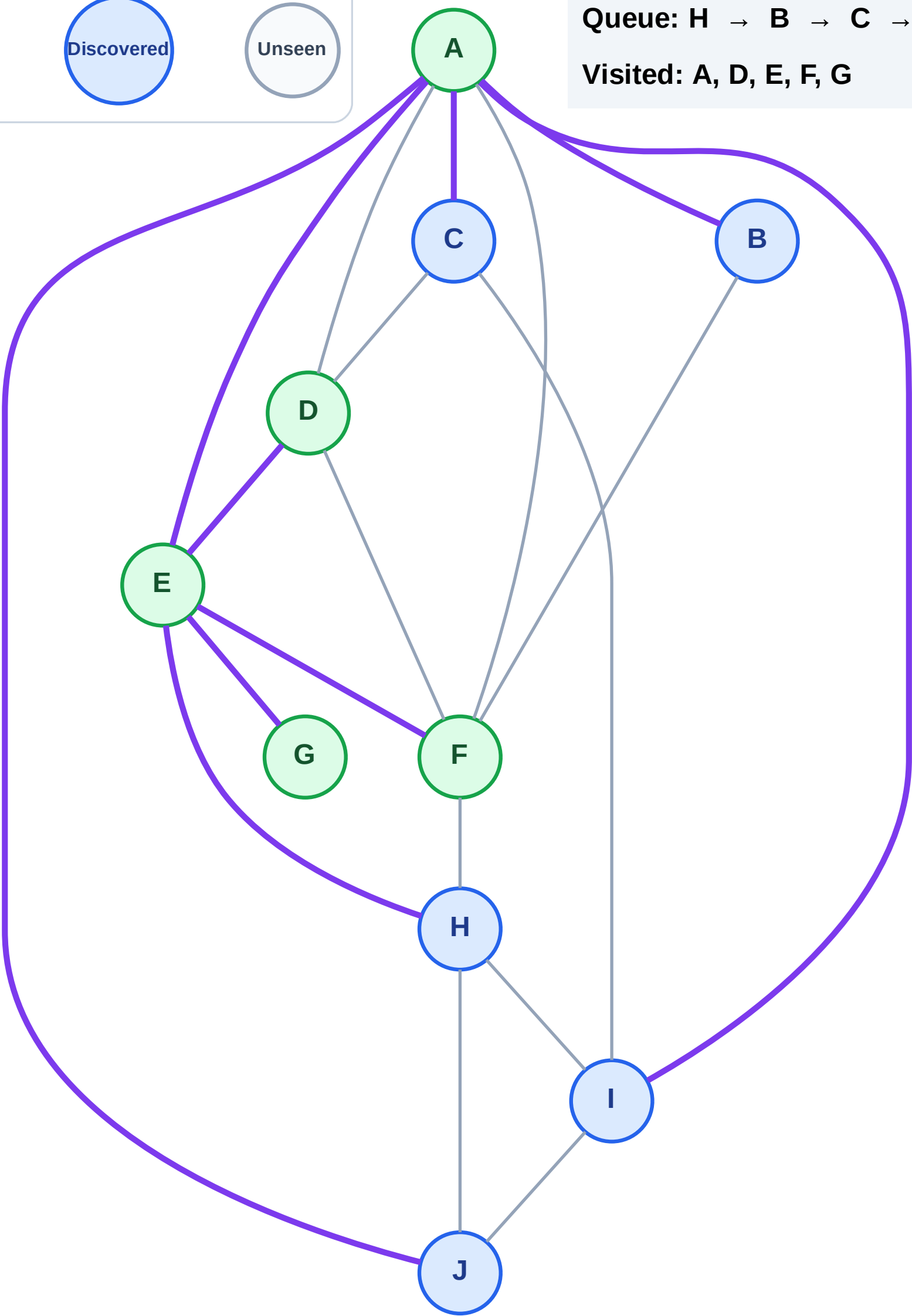
Step 11: No new neighbors from G

Legend



Queue: H → B → C → I → J

Visited: A, D, E, F, G



BFS Traversal

Step 12: Process node H

Legend

Complete

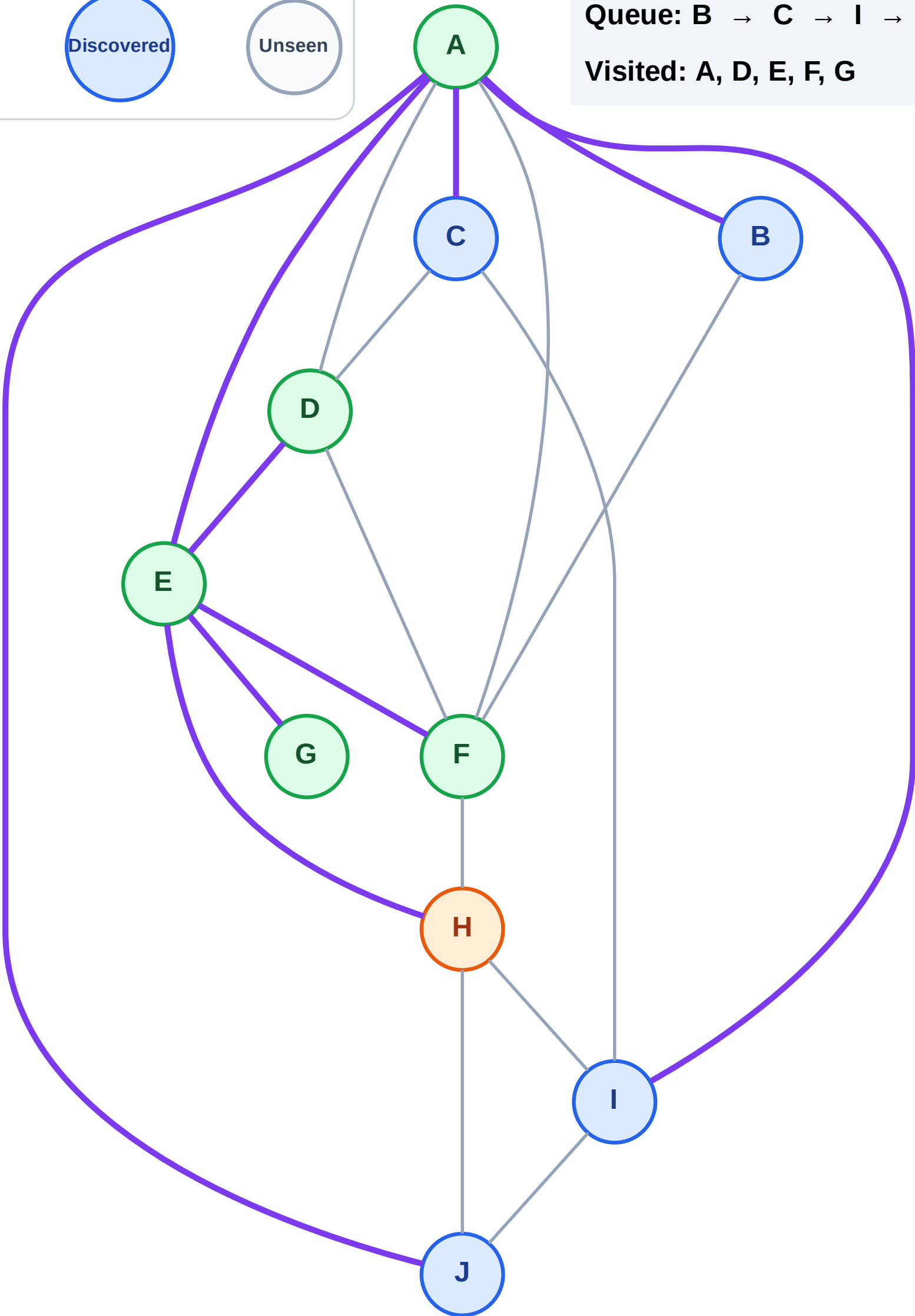
Processing

Discovered

Unseen

Queue: B → C → I → J

Visited: A, D, E, F, G



BFS Traversal

Step 13: No new neighbors from H

Legend

Complete

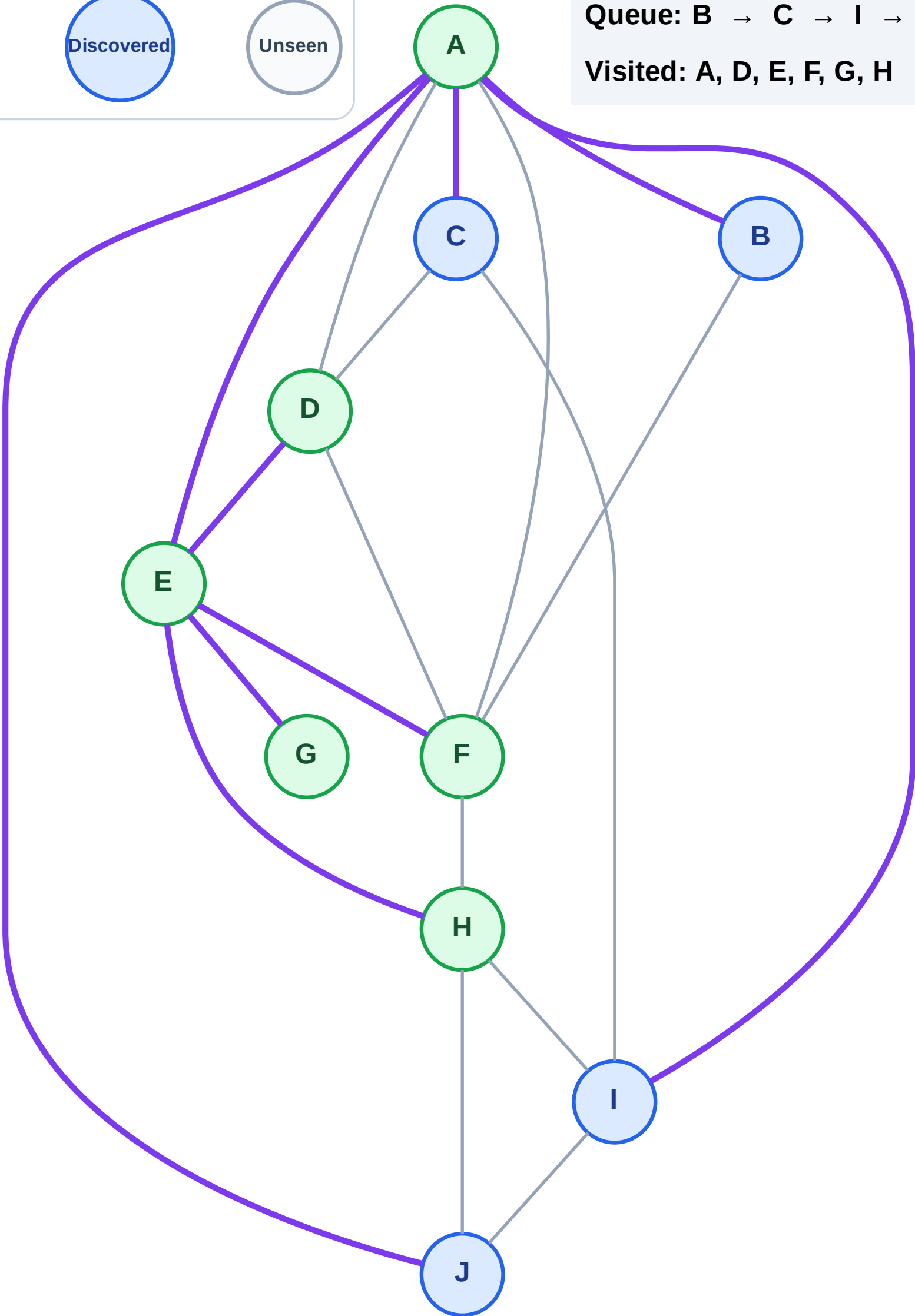
Processing

Discovered

Unseen

Queue: B → C → I → J

Visited: A, D, E, F, G, H



BFS Traversal

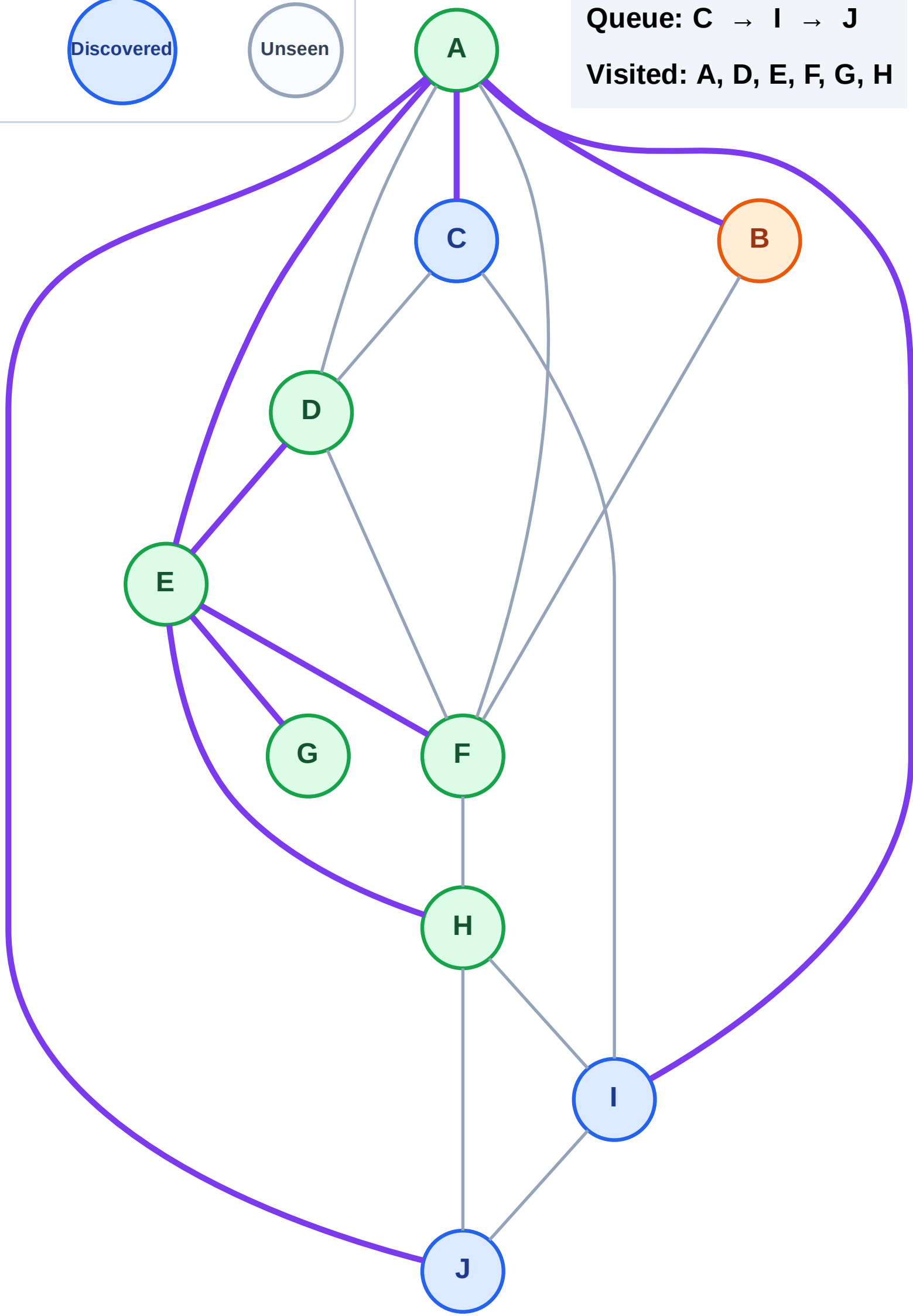
Step 14: Process node B

Legend



Queue: C → I → J

Visited: A, D, E, F, G, H



BFS Traversal

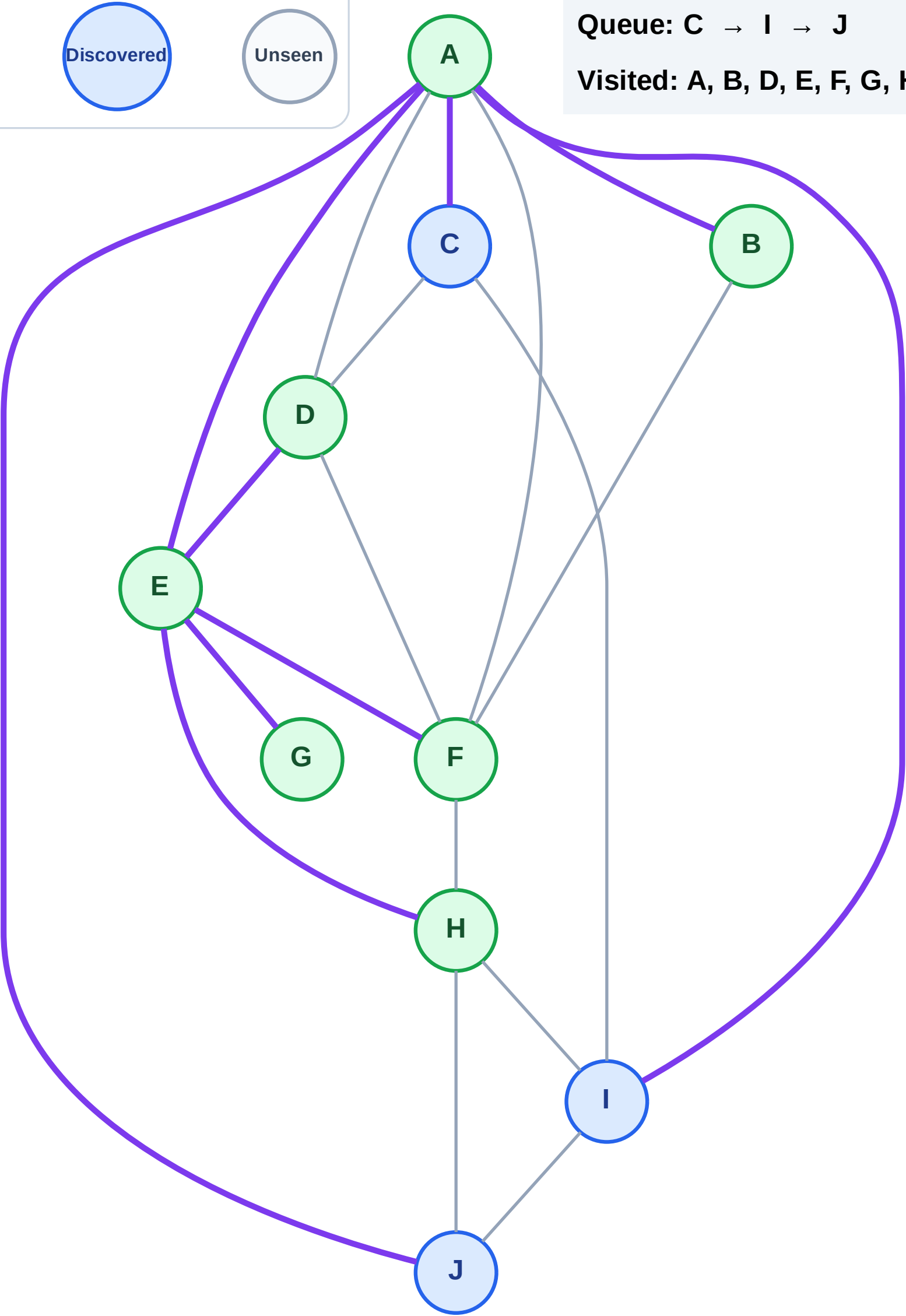
Step 15: No new neighbors from B

Legend



Queue: C → I → J

Visited: A, B, D, E, F, G, H



BFS Traversal

Step 16: Process node C

Legend

Complete

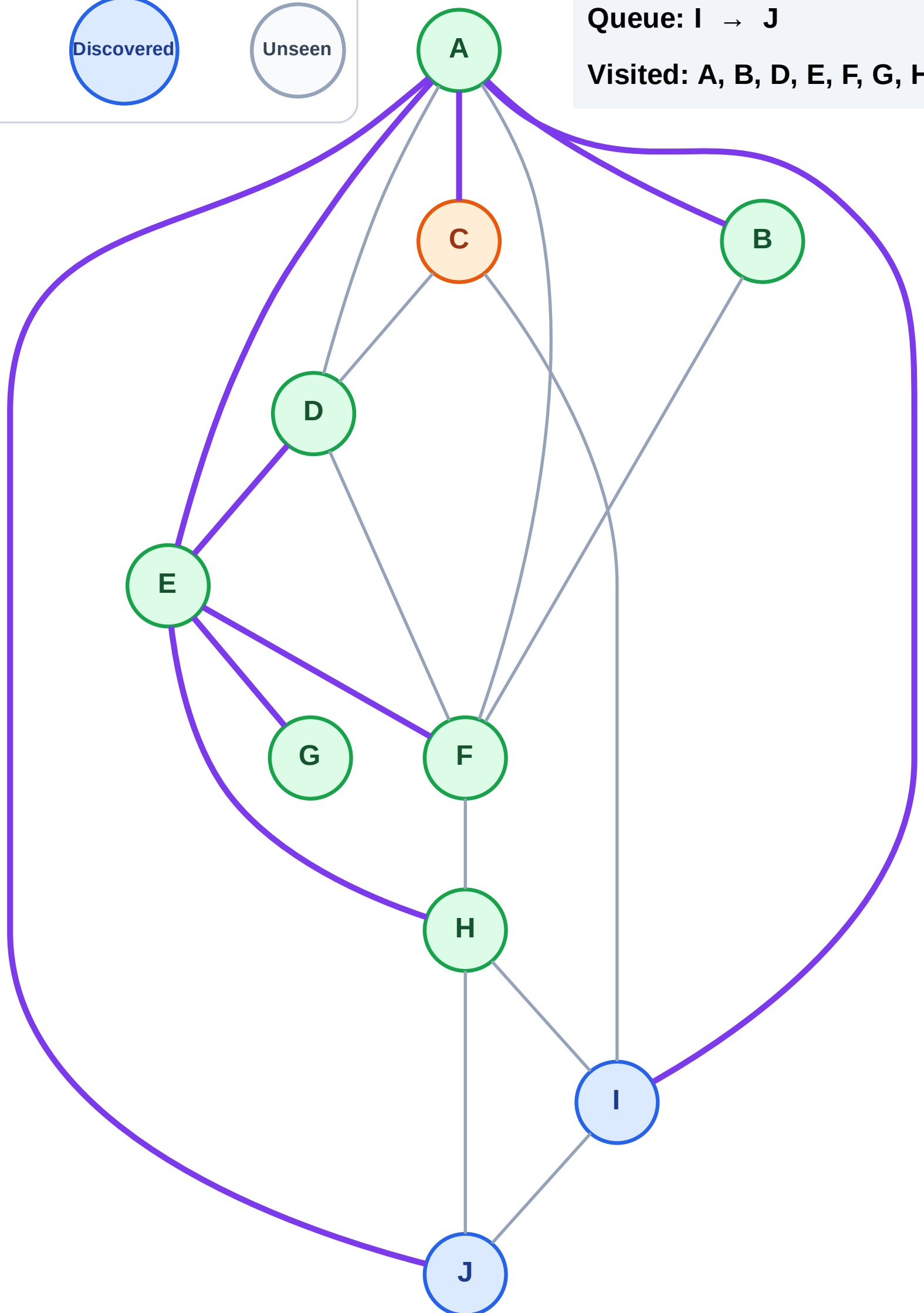
Processing

Discovered

Unseen

Queue: I → J

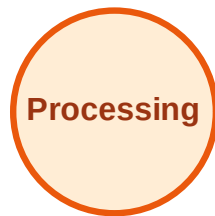
Visited: A, B, D, E, F, G, H



BFS Traversal

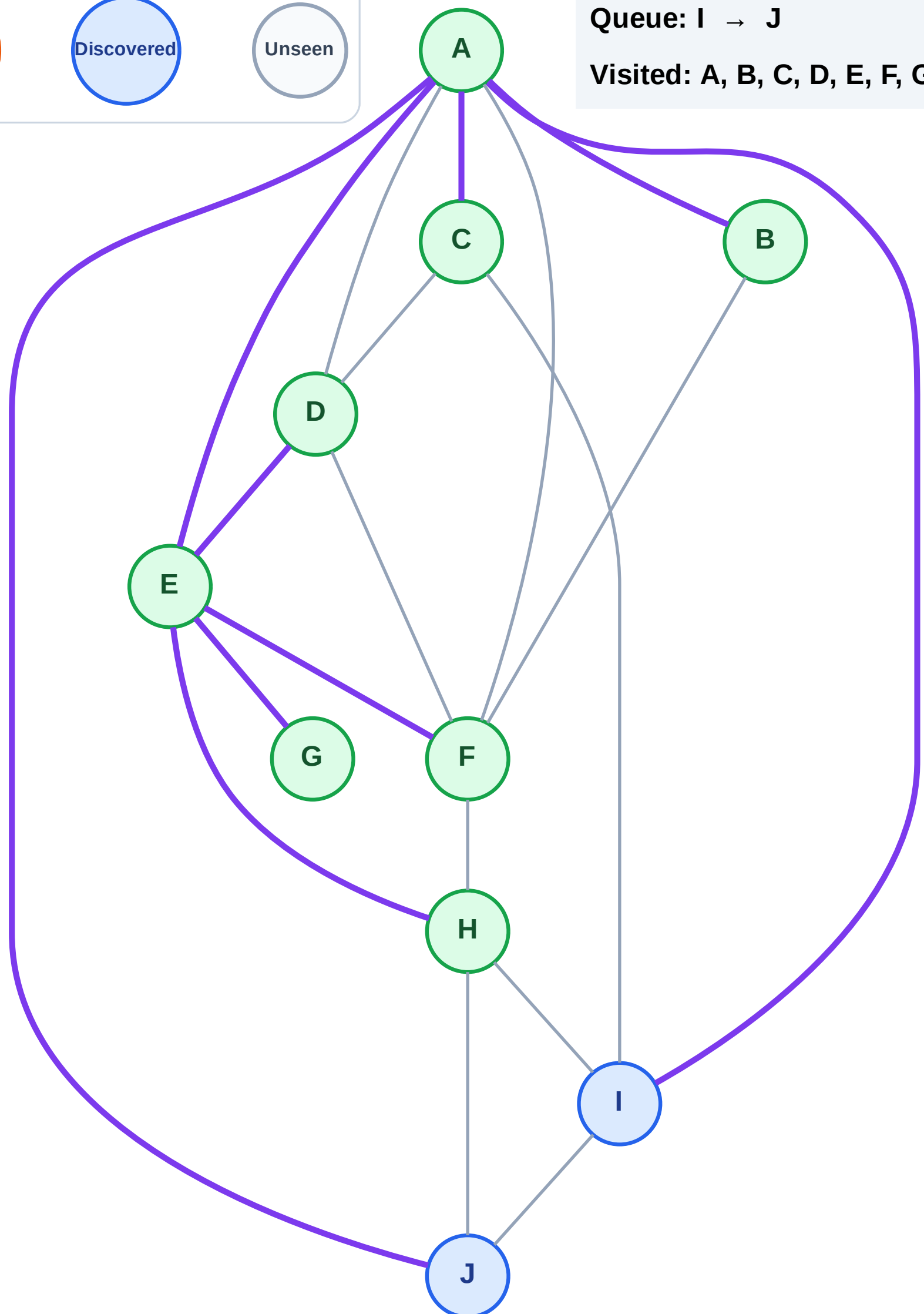
Step 17: No new neighbors from C

Legend



Queue: I → J

Visited: A, B, C, D, E, F, G, H



BFS Traversal

Step 18: Process node I

Legend

Complete

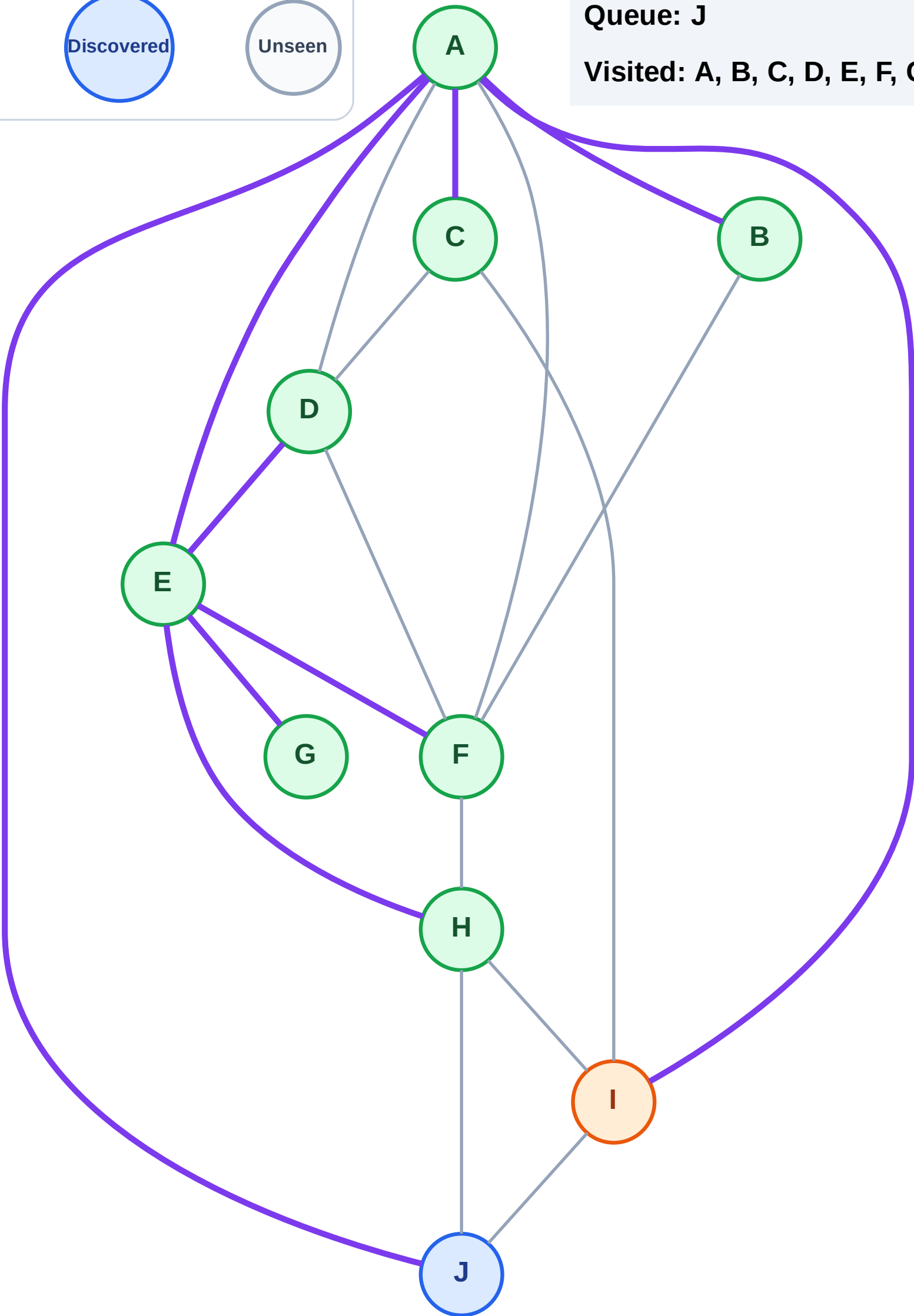
Processing

Discovered

Unseen

Queue: J

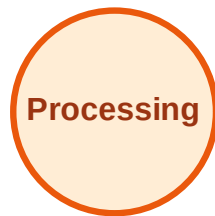
Visited: A, B, C, D, E, F, G, H



BFS Traversal

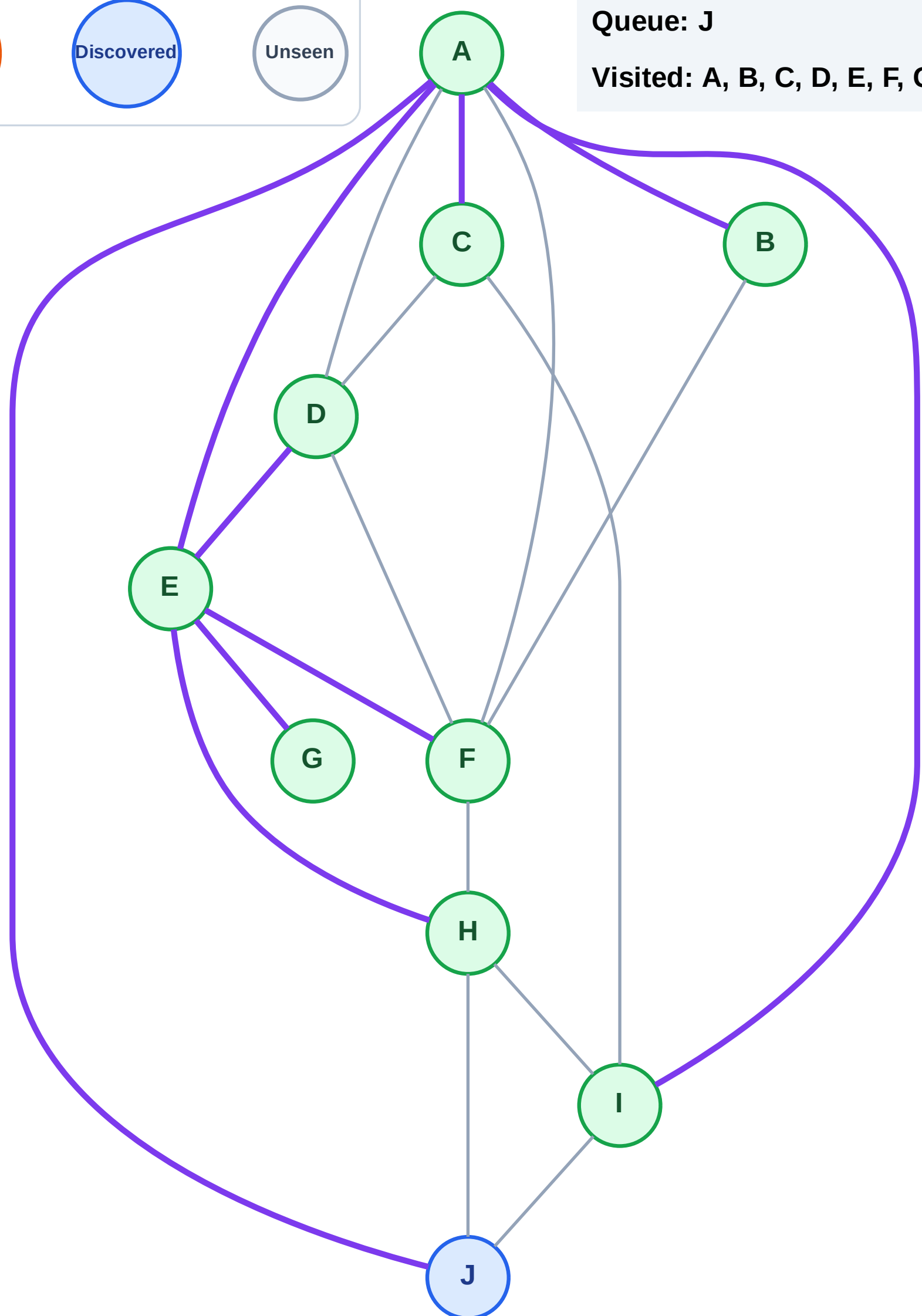
Step 19: No new neighbors from I

Legend



Queue: J

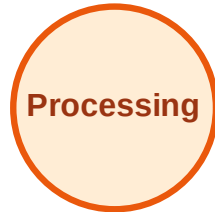
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

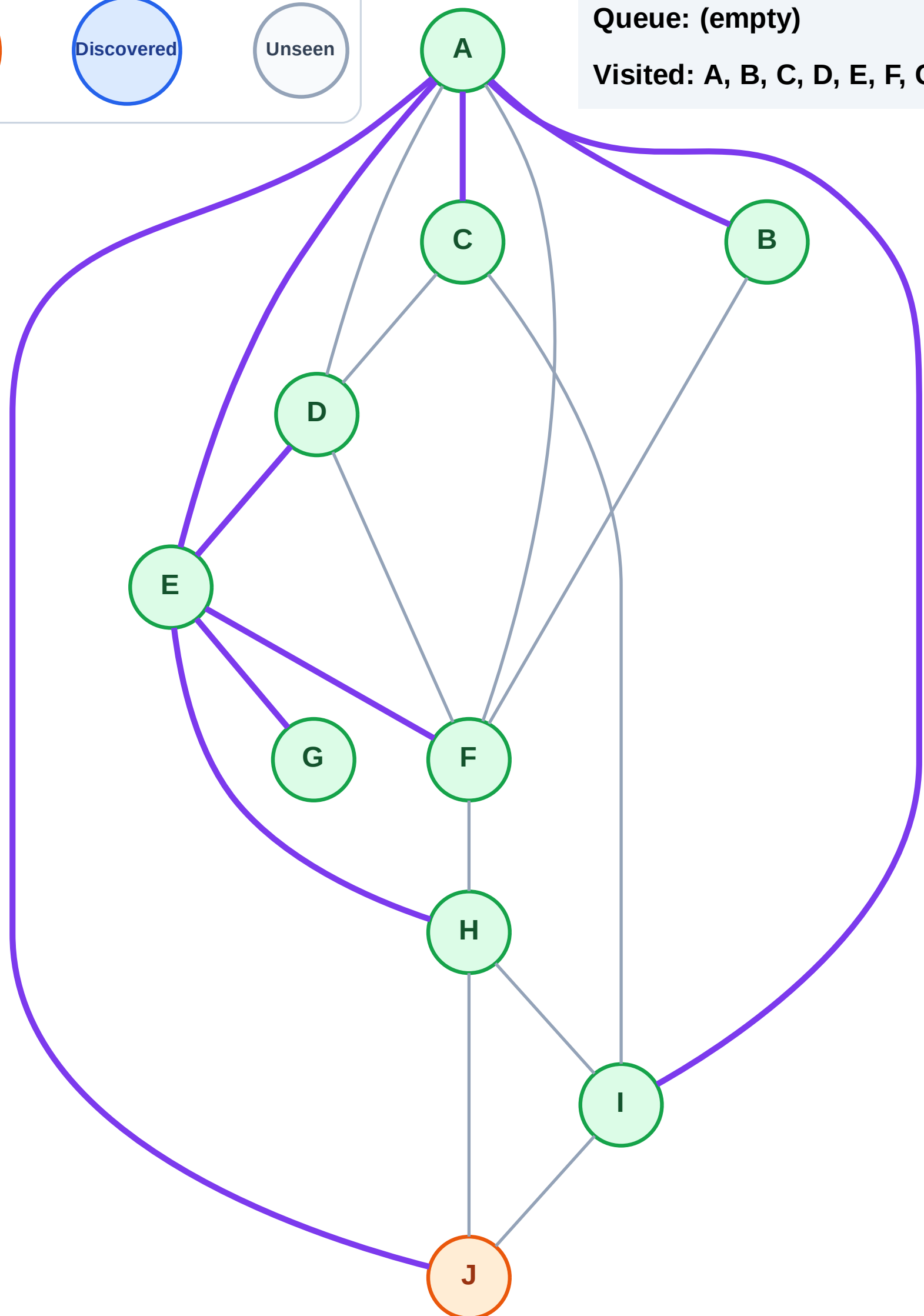
Step 20: Process node J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

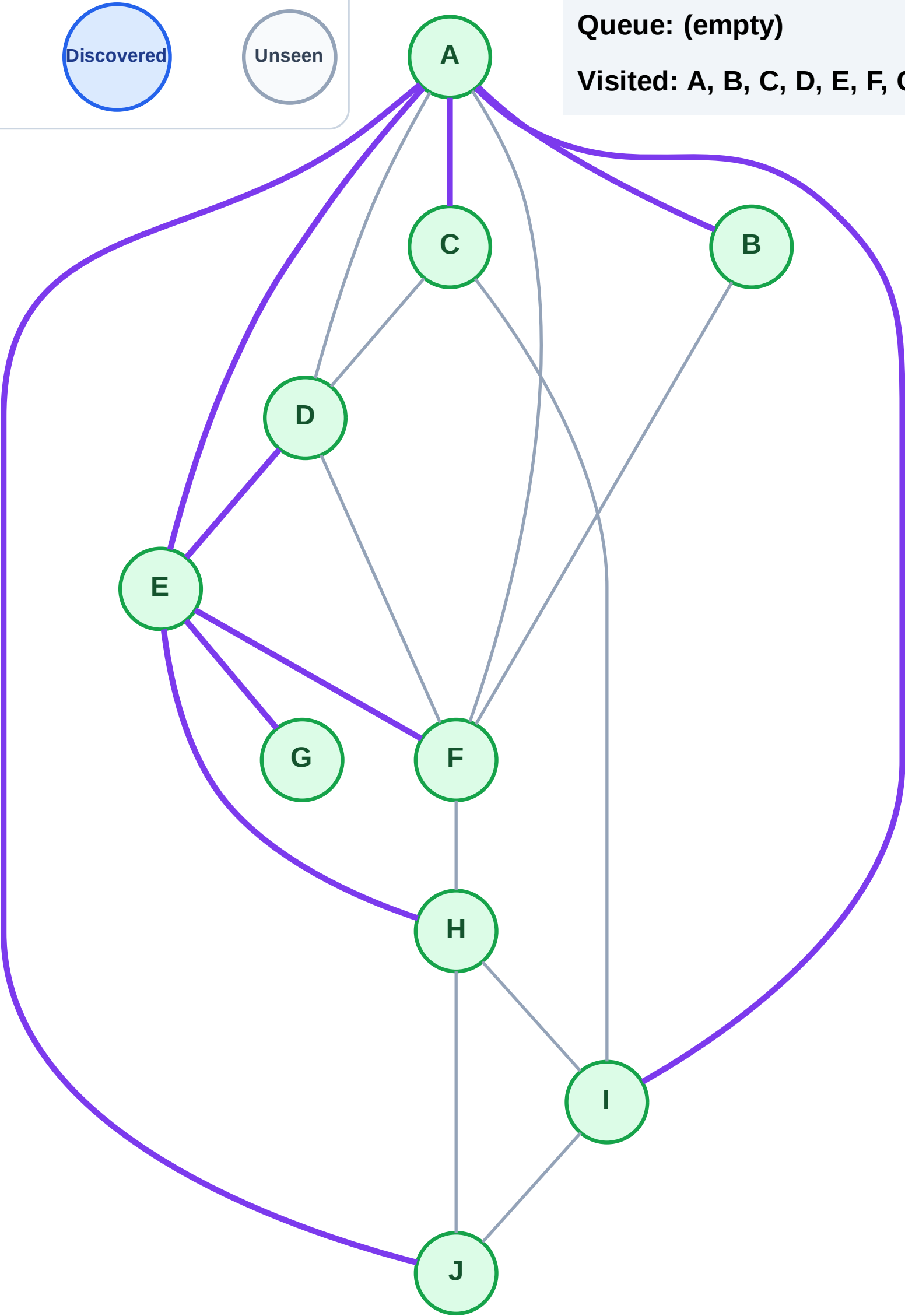
Step 21: No new neighbors from J

Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

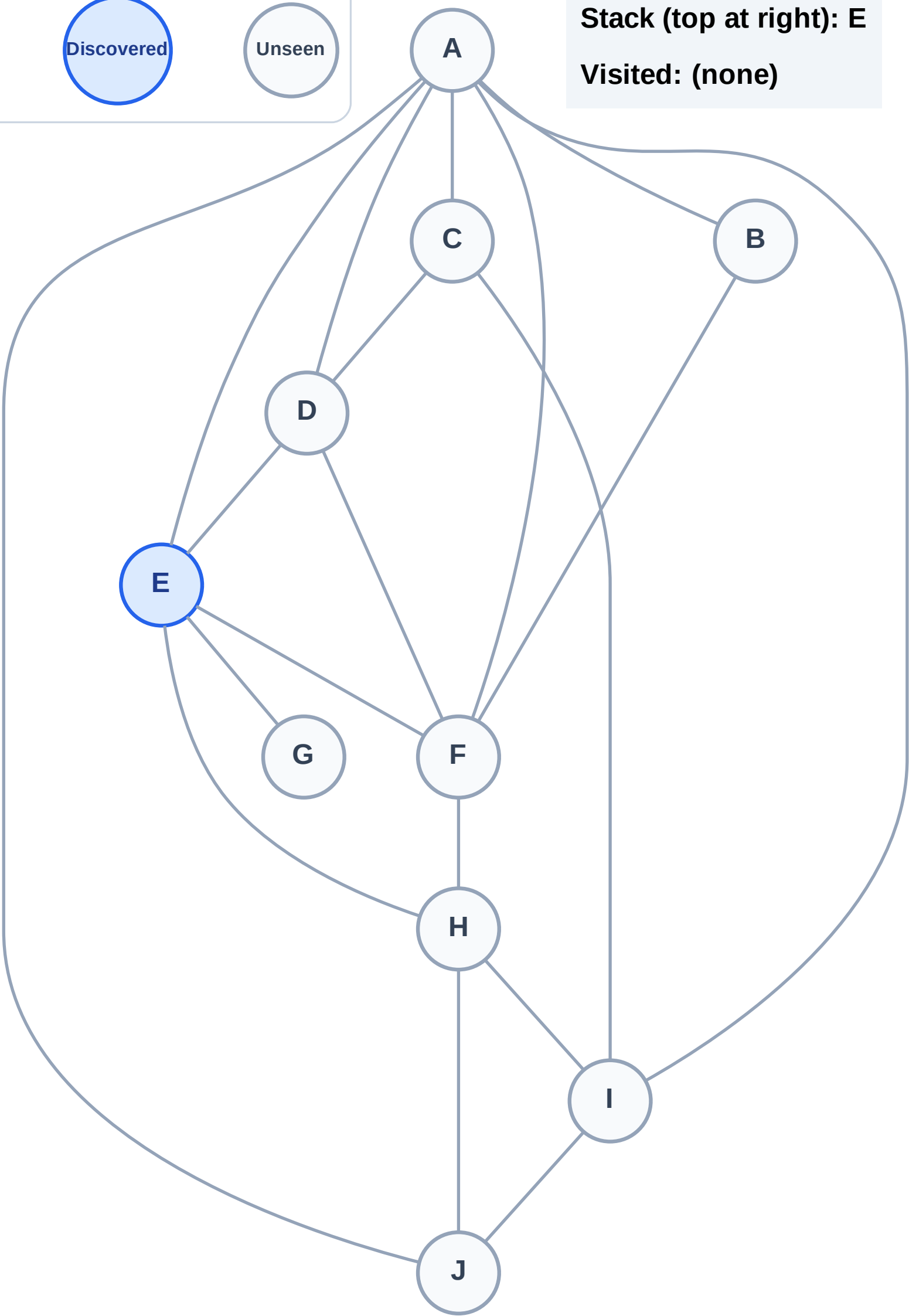
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

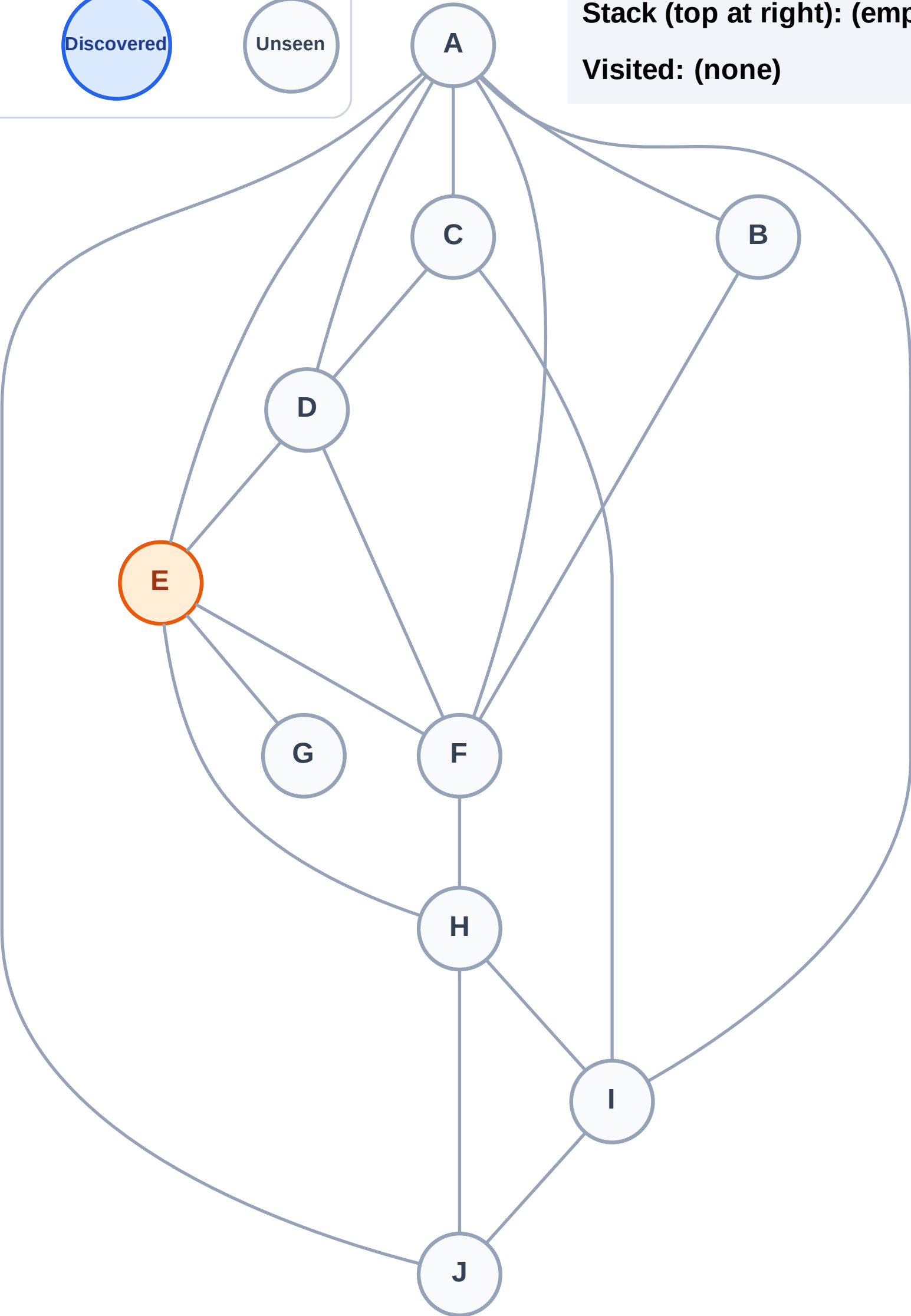
Complete

Processing

Discovered

Unseen

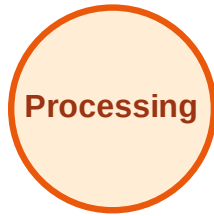
Stack (top at right): (empty)
Visited: (none)



DFS Traversal

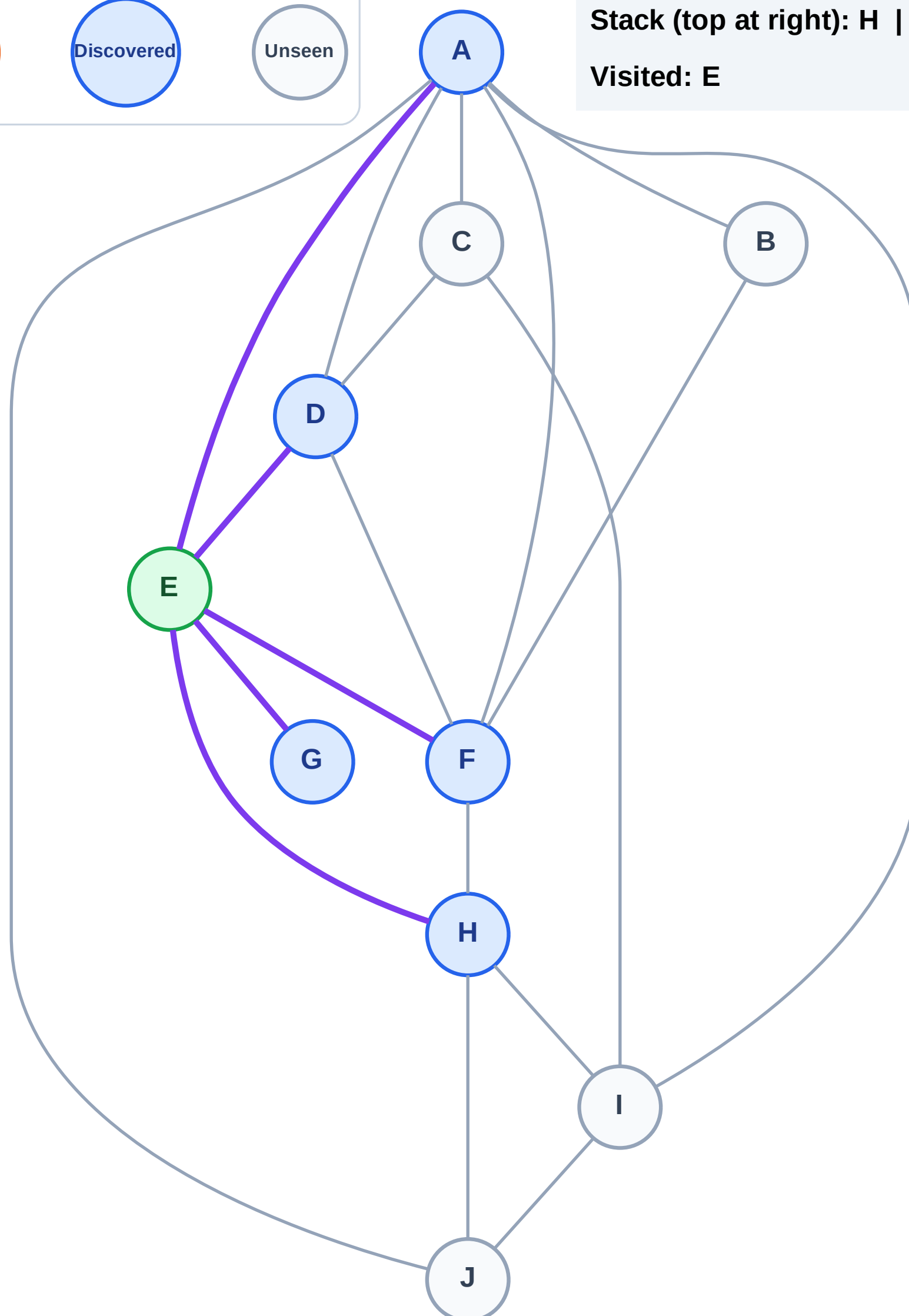
Step 3: Discover H, G, F, D, A from E

Legend



Stack (top at right): H | G | F | D | A

Visited: E



DFS Traversal

Step 4: Process node A

Legend

Complete

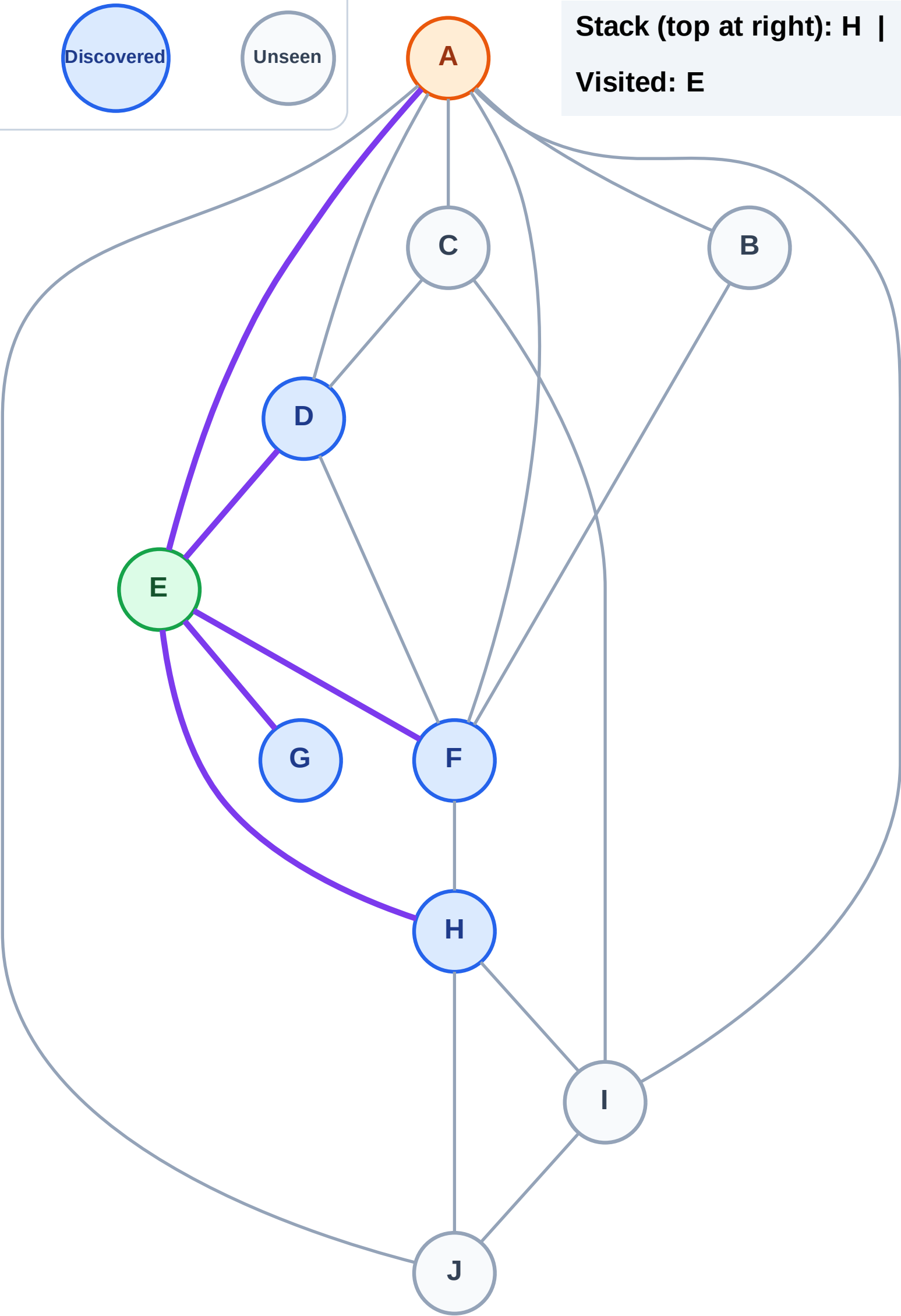
Processing

Discovered

Unseen

Stack (top at right): H | G | F | D

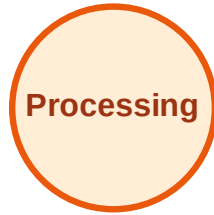
Visited: E



DFS Traversal

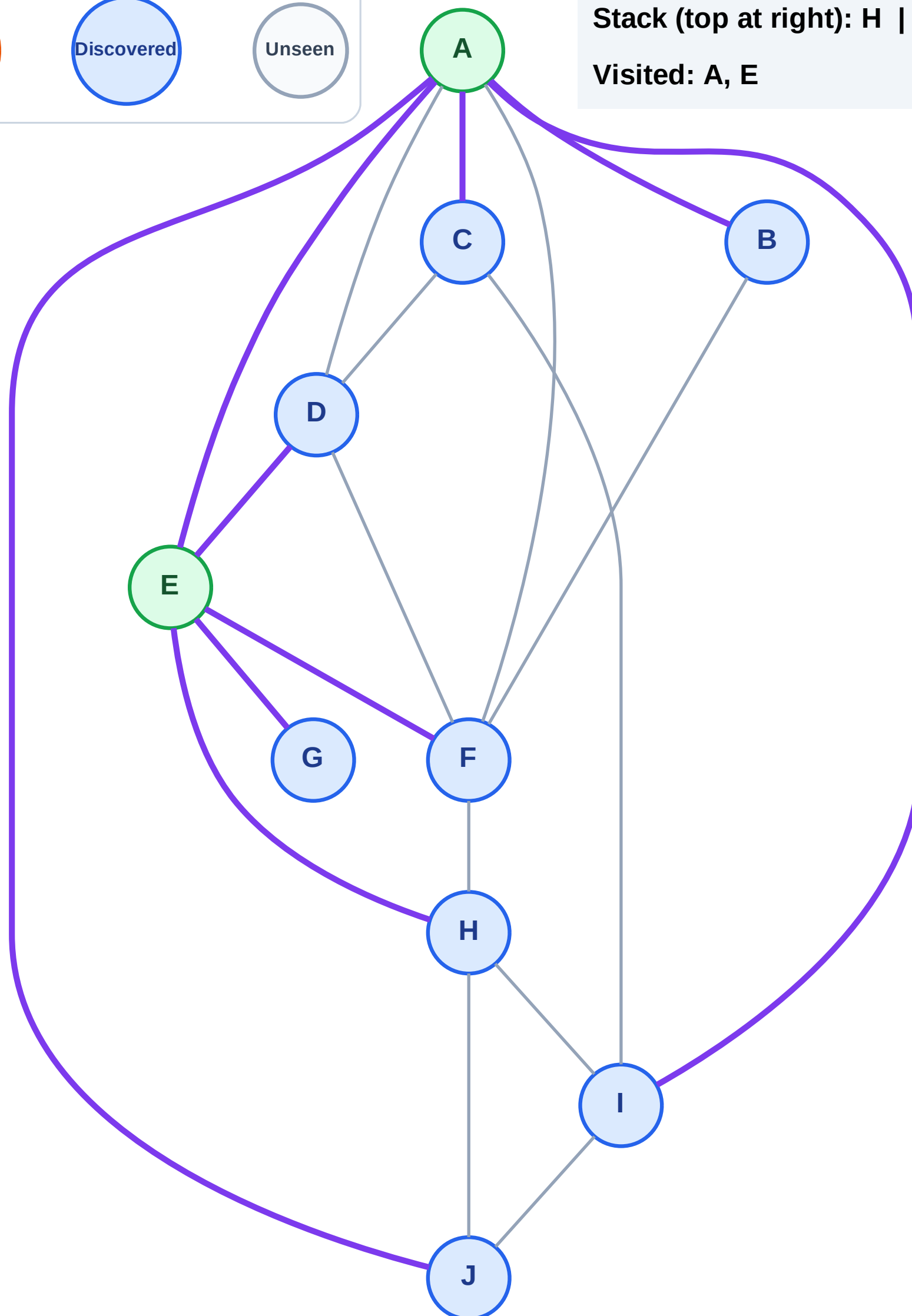
Step 5: Discover J, I, C, B from A

Legend



Stack (top at right): H | G | F | D | J | I | C | B

Visited: A, E



DFS Traversal

Step 6: Process node B

Legend

Complete

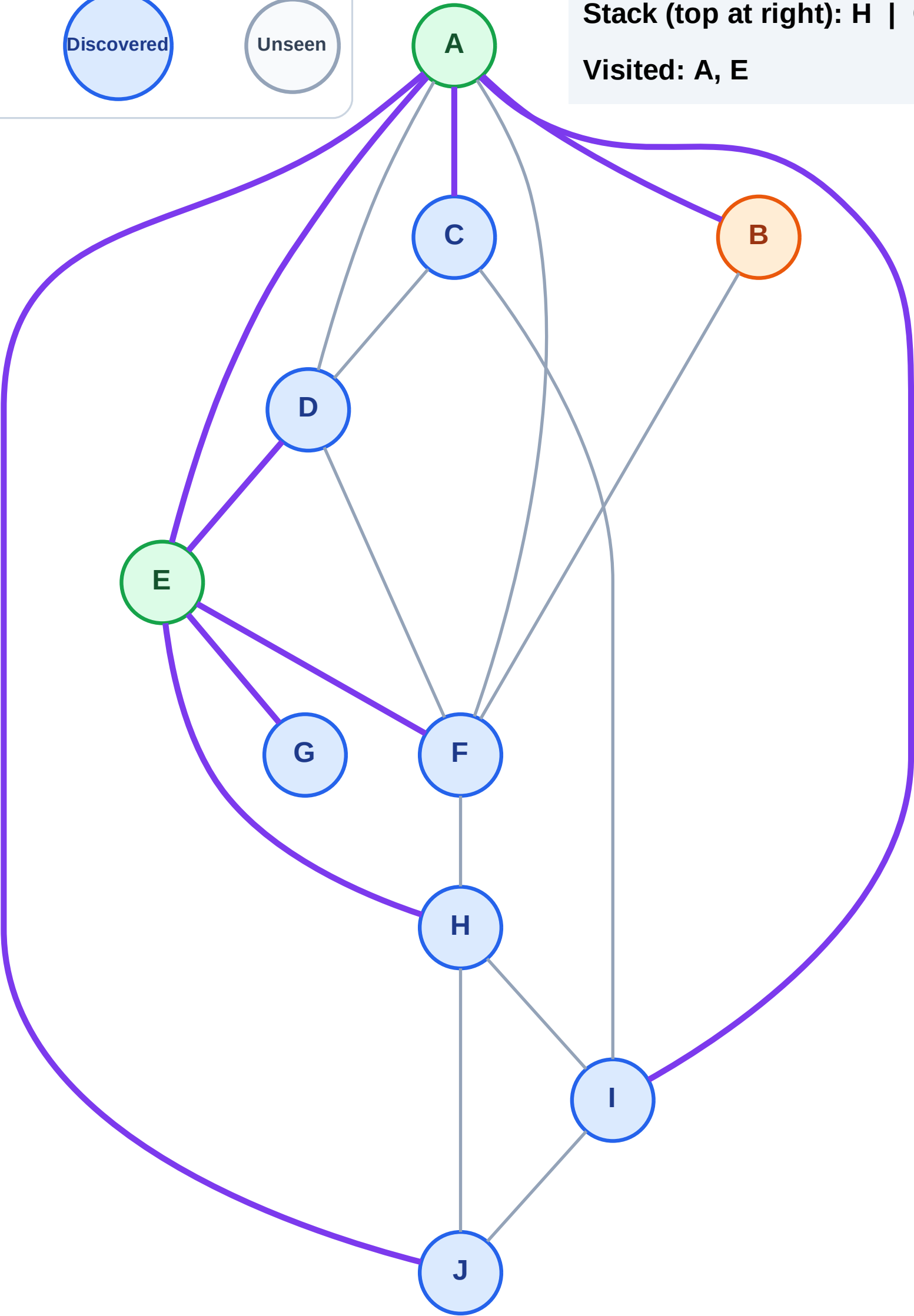
Processing

Discovered

Unseen

Stack (top at right): H | G | F | D | J | I | C

Visited: A, E



DFS Traversal

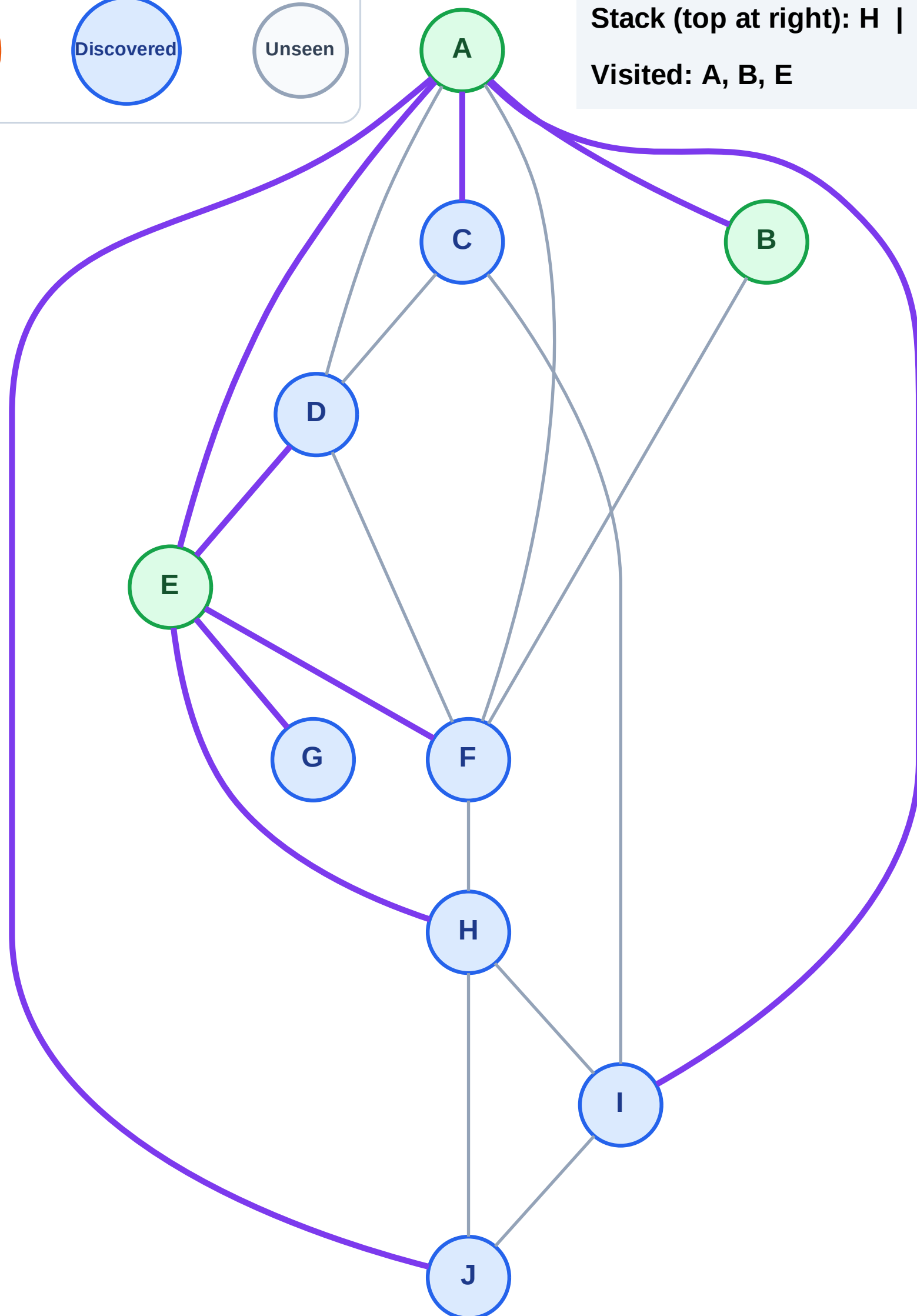
Step 7: No new neighbors from B

Legend



Stack (top at right): H | G | F | D | J | I | C

Visited: A, B, E



DFS Traversal

Step 8: Process node C

Legend



Complete



Processing



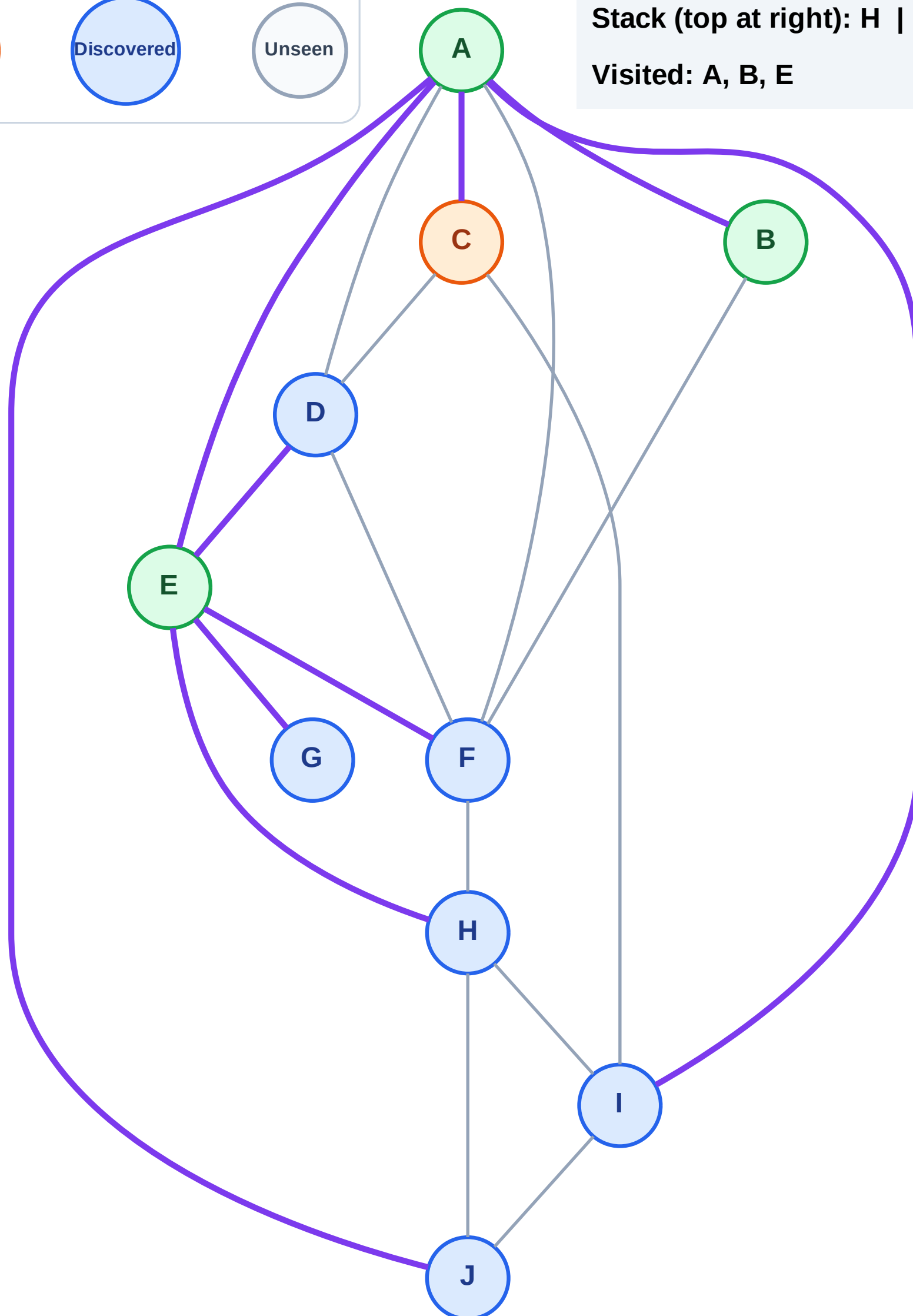
Discovered



Unseen

Stack (top at right): H | G | F | D | J | I

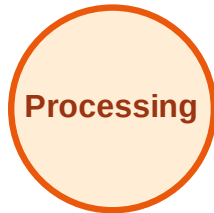
Visited: A, B, E



DFS Traversal

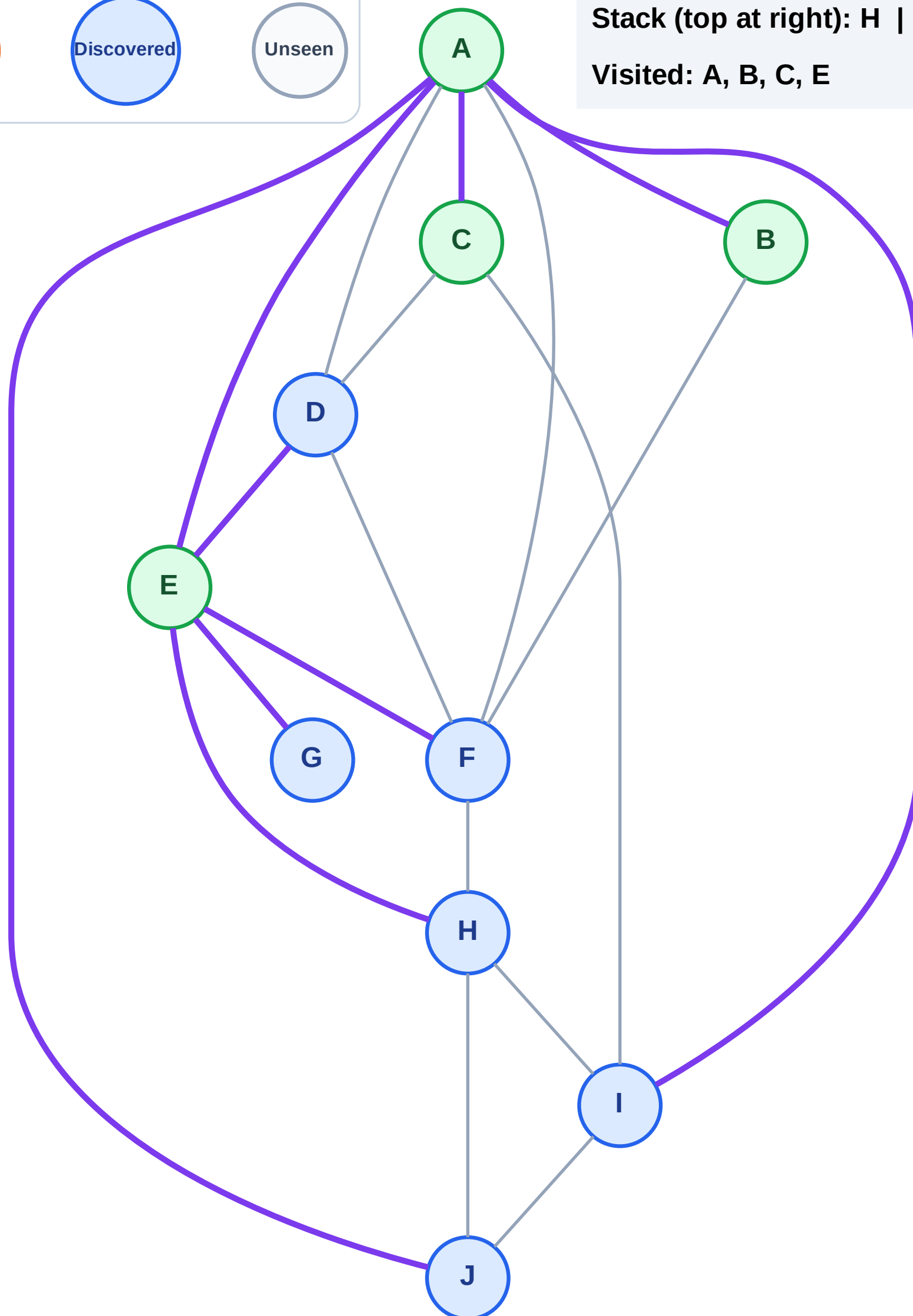
Step 9: No new neighbors from C

Legend



Stack (top at right): H | G | F | D | J | I

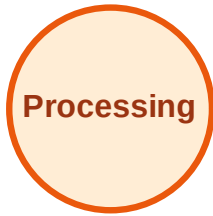
Visited: A, B, C, E



DFS Traversal

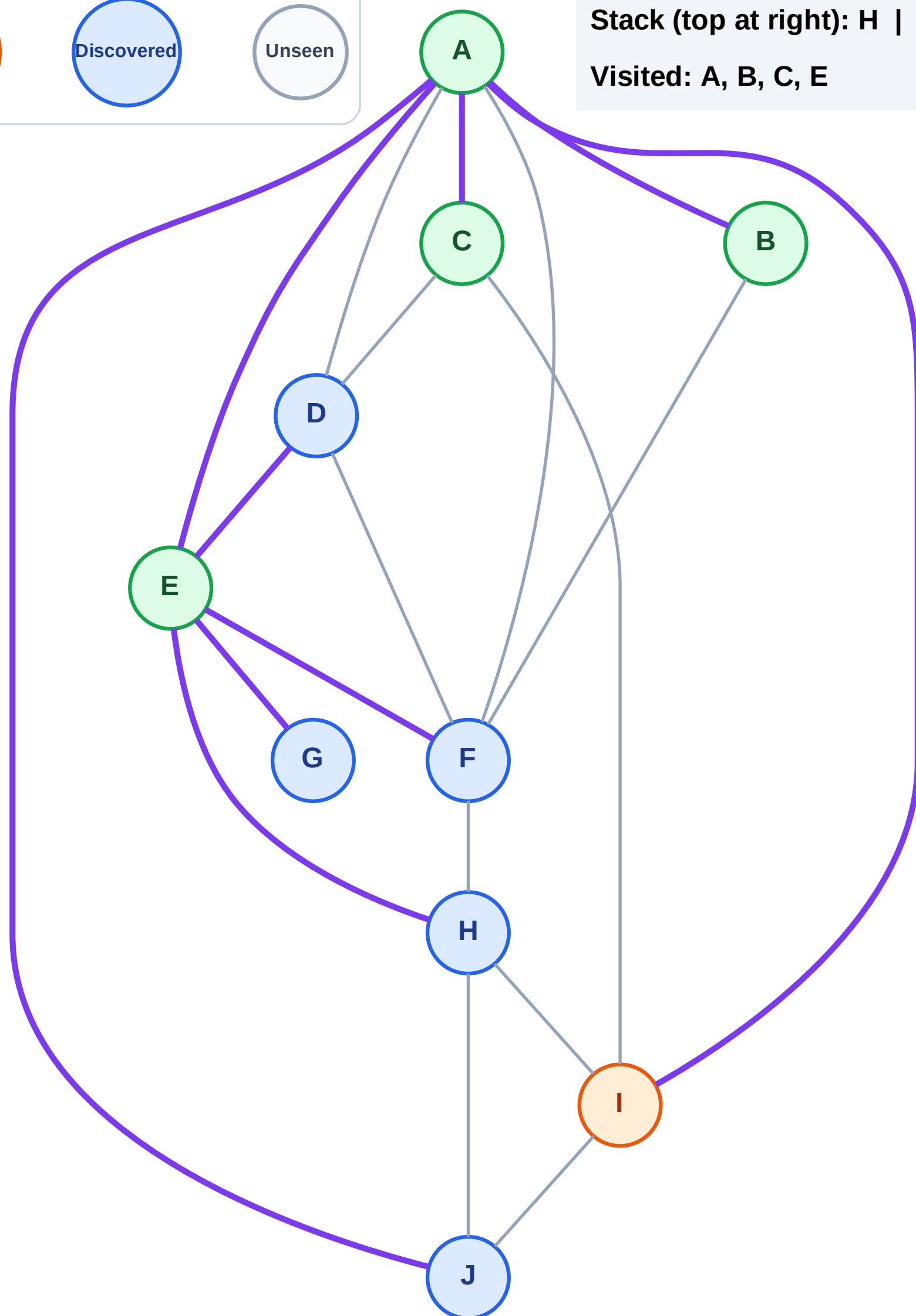
Step 10: Process node I

Legend



Stack (top at right): H | G | F | D | J

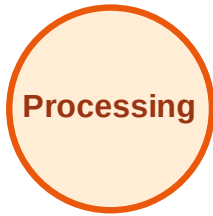
Visited: A, B, C, E



DFS Traversal

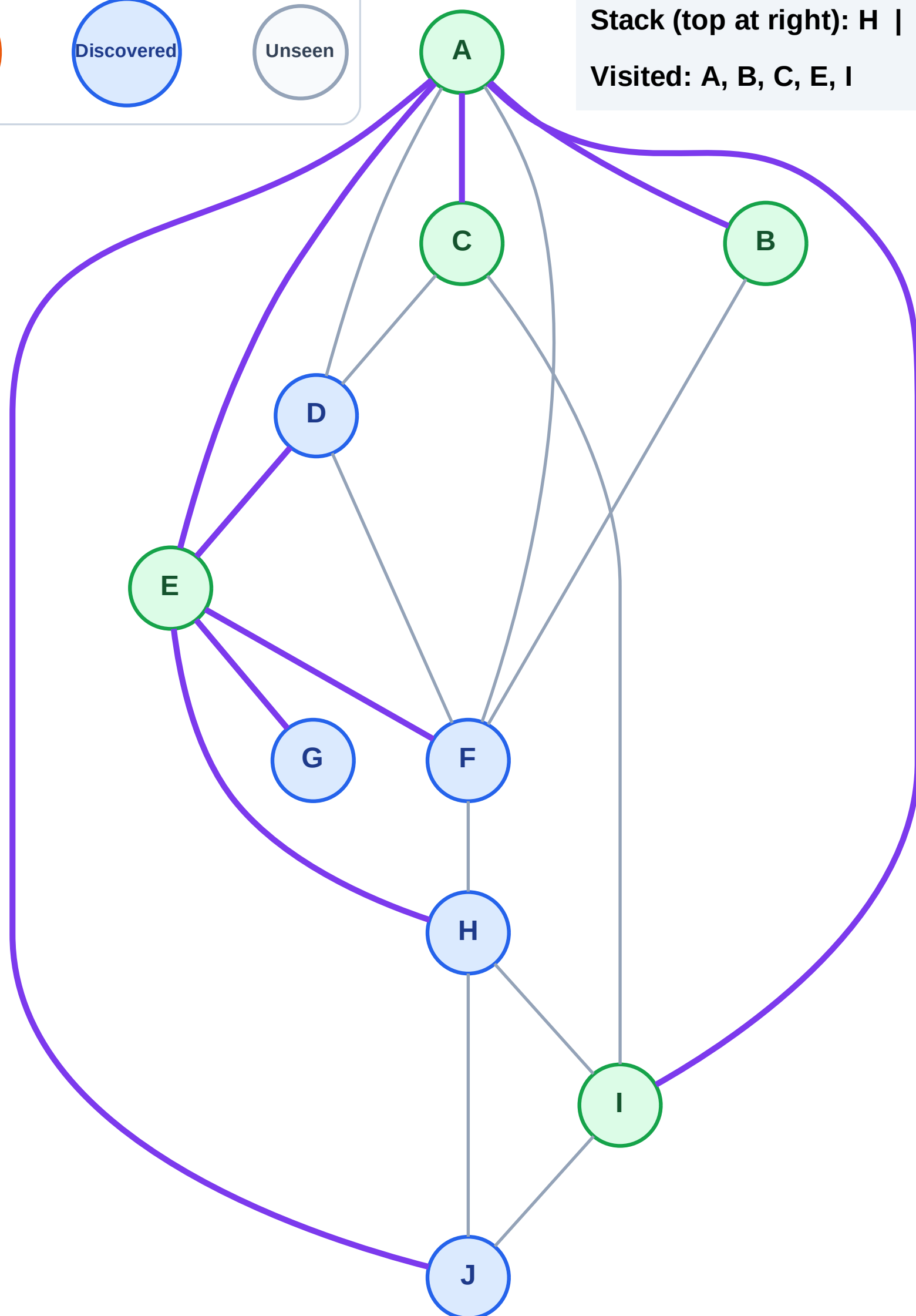
Step 11: No new neighbors from I

Legend



Stack (top at right): H | G | F | D | J

Visited: A, B, C, E, I



DFS Traversal

Step 12: Process node J

Legend

Complete

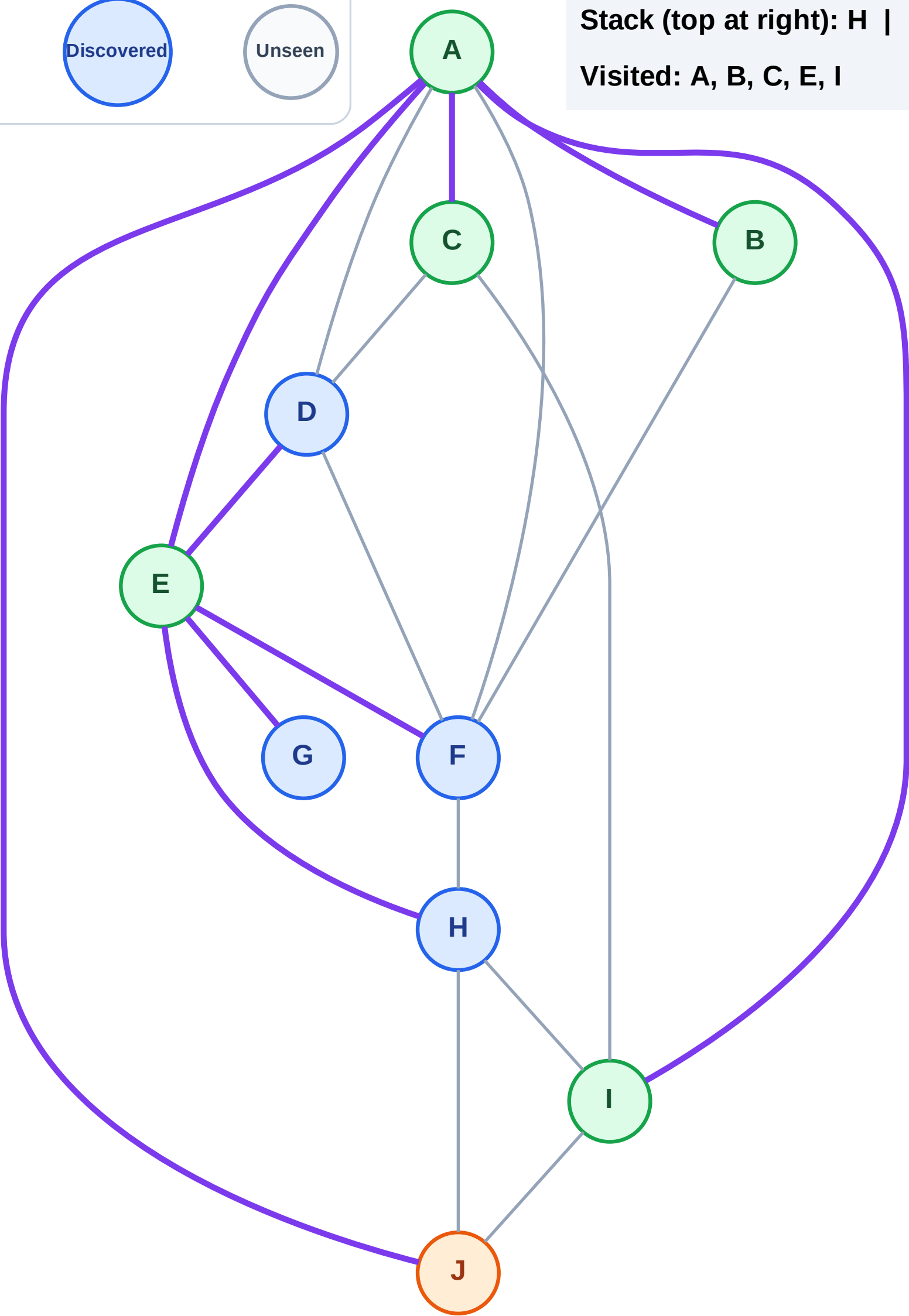
Processing

Discovered

Unseen

Stack (top at right): H | G | F | D

Visited: A, B, C, E, I



DFS Traversal

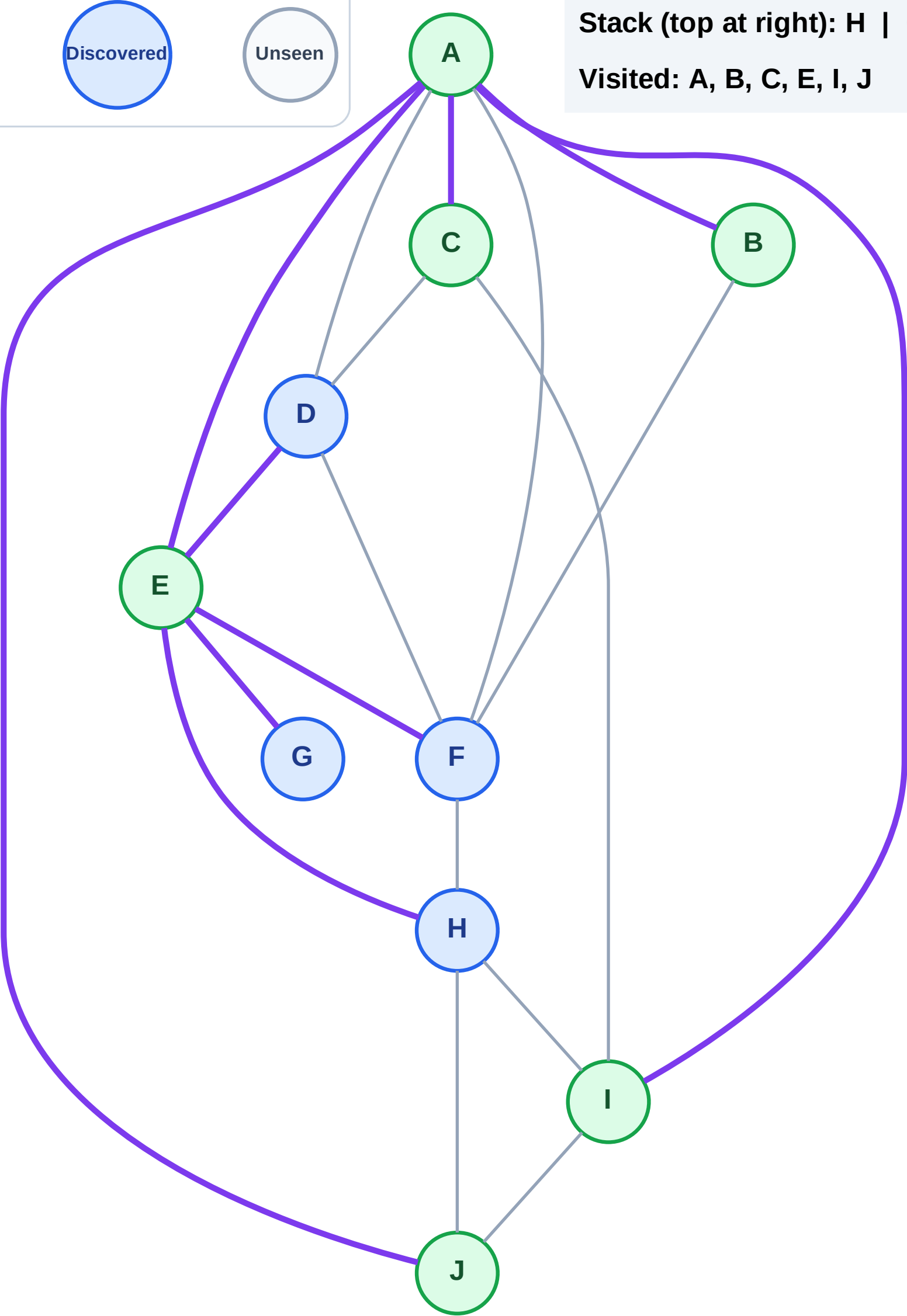
Step 13: No new neighbors from J

Legend



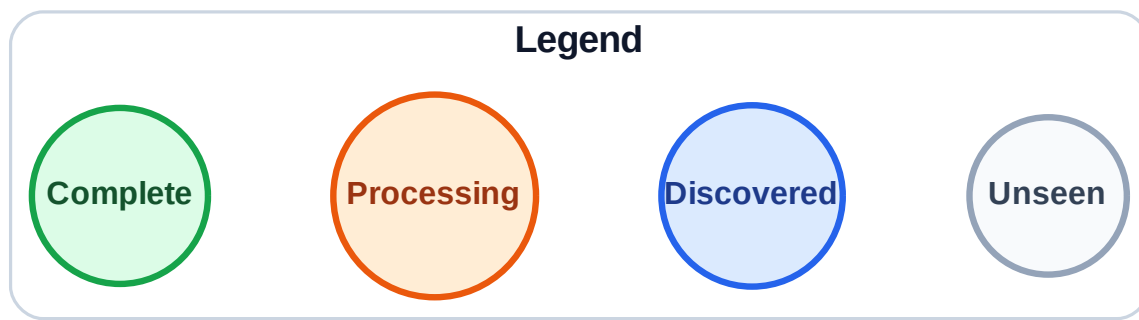
Stack (top at right): H | G | F | D

Visited: A, B, C, E, I, J

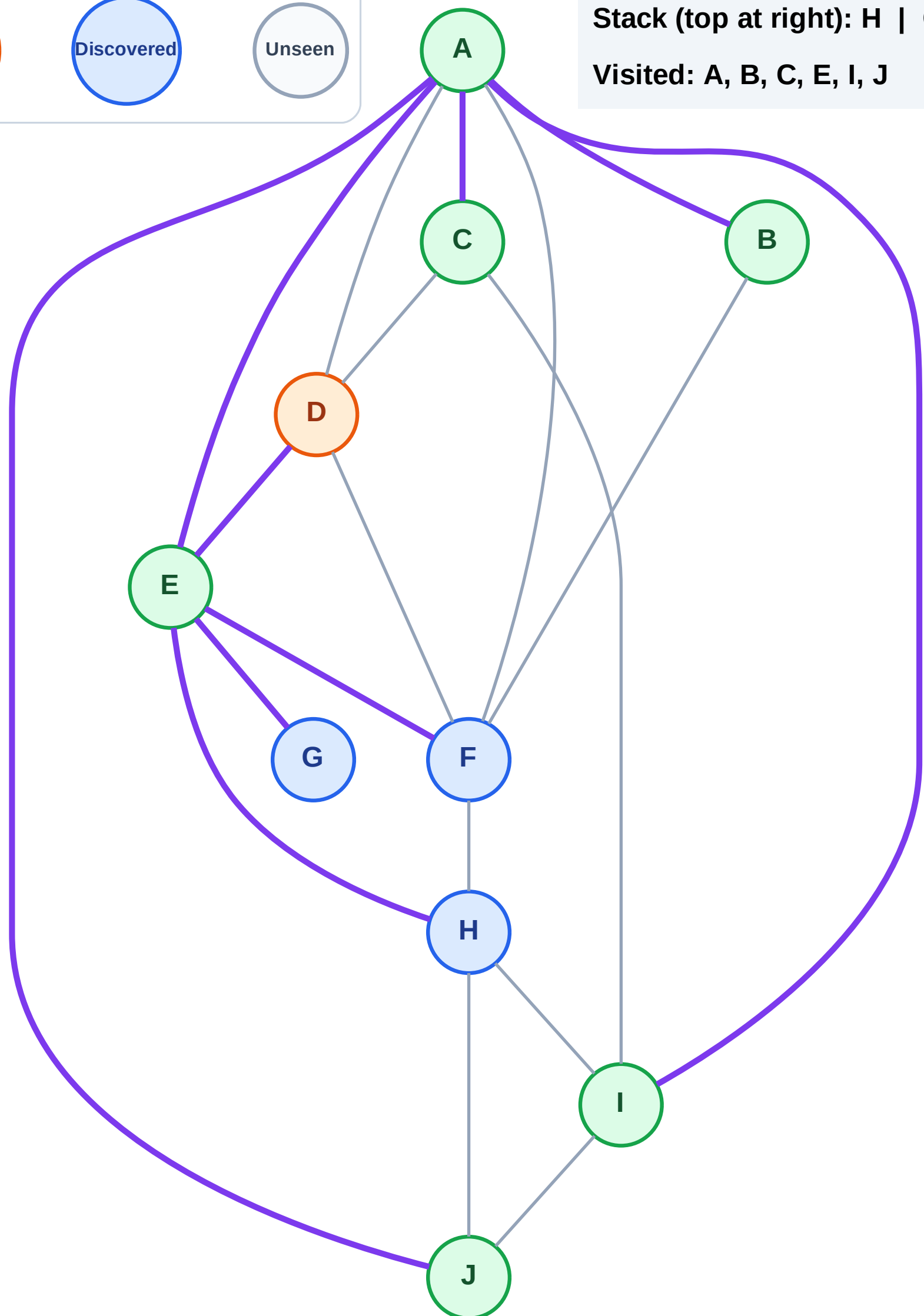


DFS Traversal

Step 14: Process node D



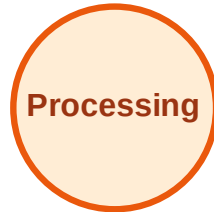
Stack (top at right): H | G | F
Visited: A, B, C, E, I, J



DFS Traversal

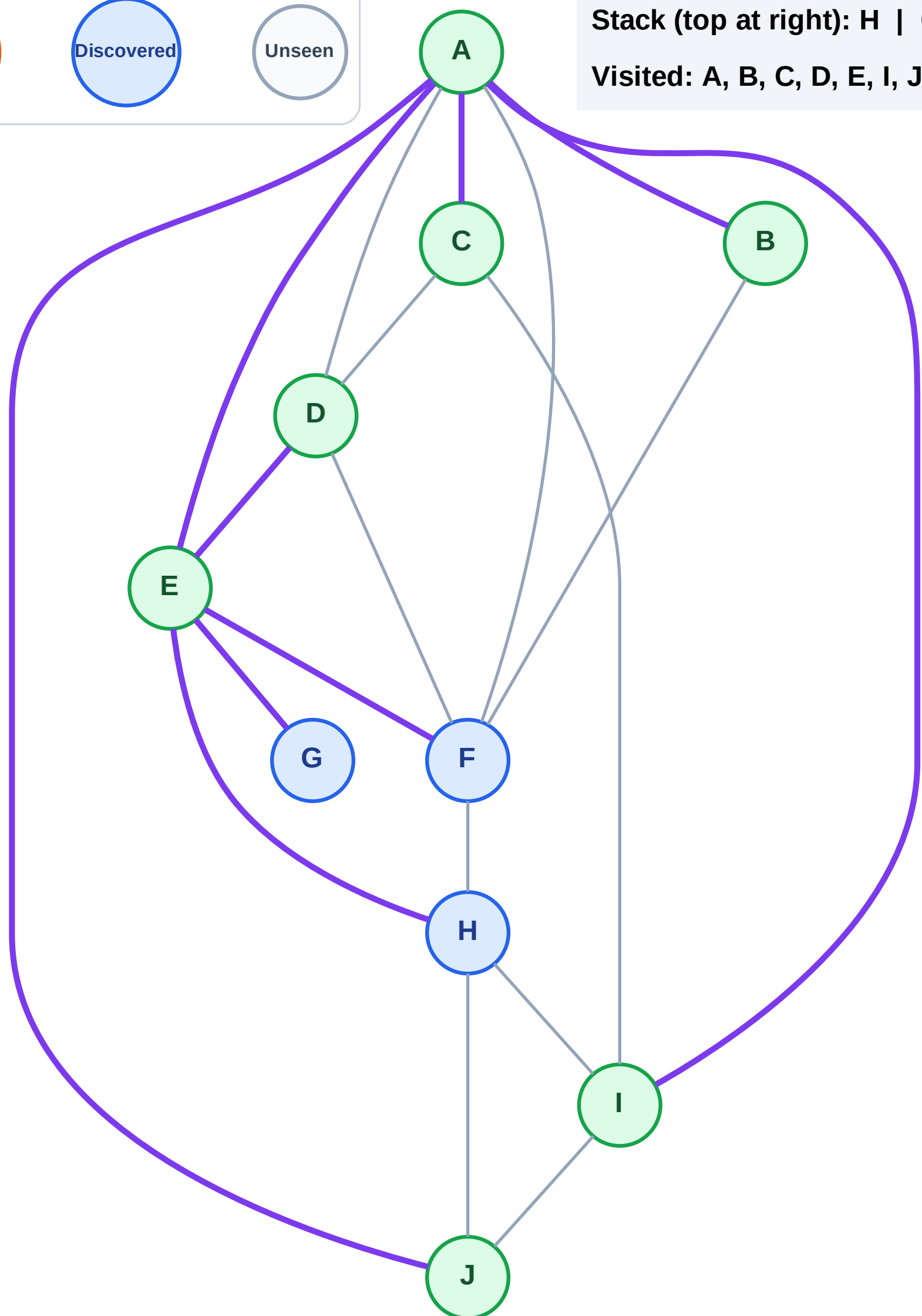
Step 15: No new neighbors from D

Legend



Stack (top at right): H | G | F

Visited: A, B, C, D, E, I, J



DFS Traversal

Step 16: Process node F

Legend

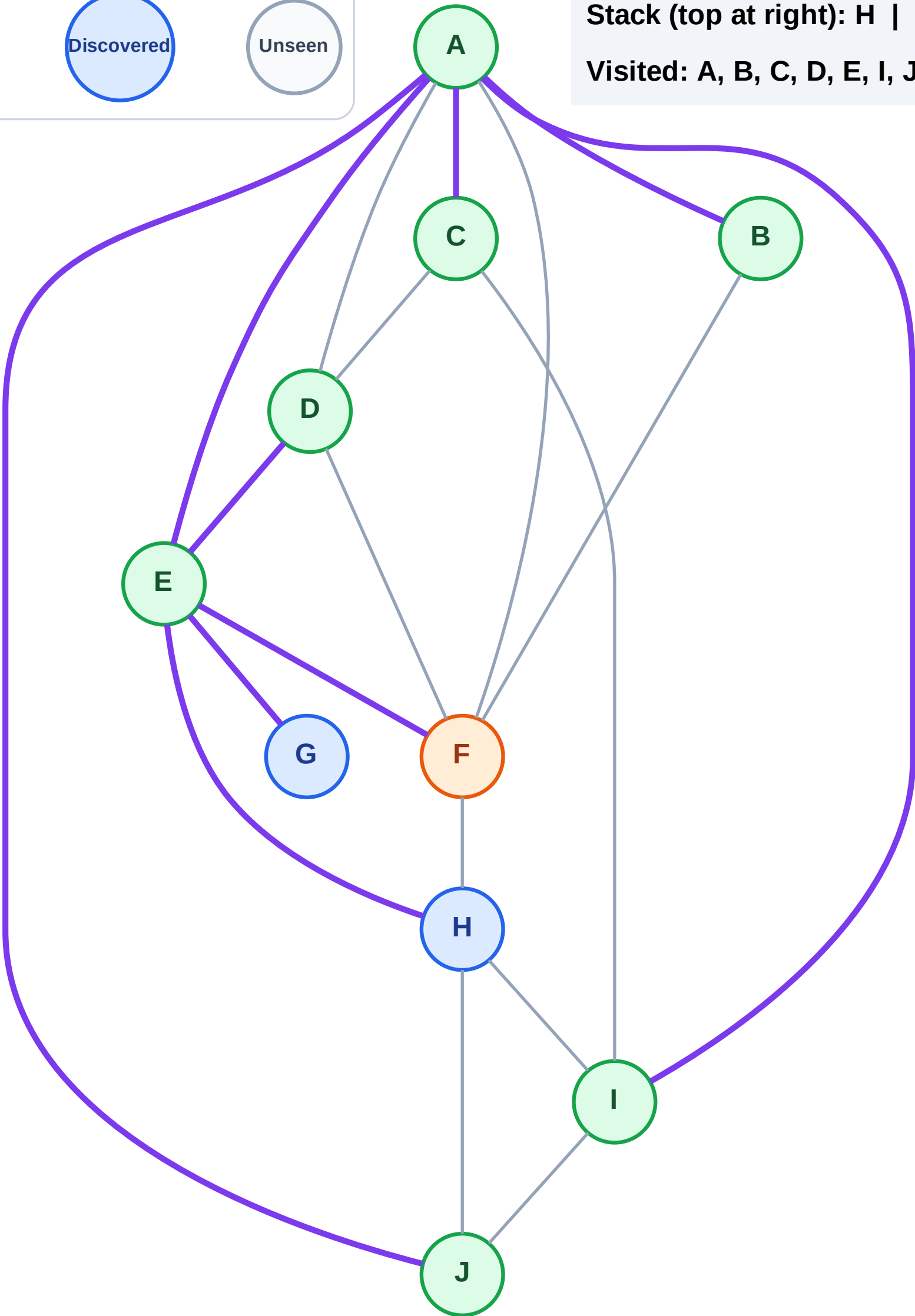
Complete

Processing

Discovered

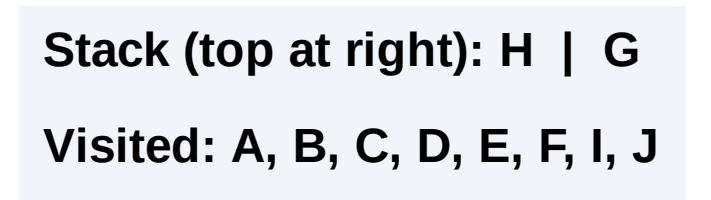
Unseen

Stack (top at right): H | G
Visited: A, B, C, D, E, I, J



Step 17: No new neighbors from F

Step 17: No new neighbors from F



DFS Traversal

Step 18: Process node G

Legend

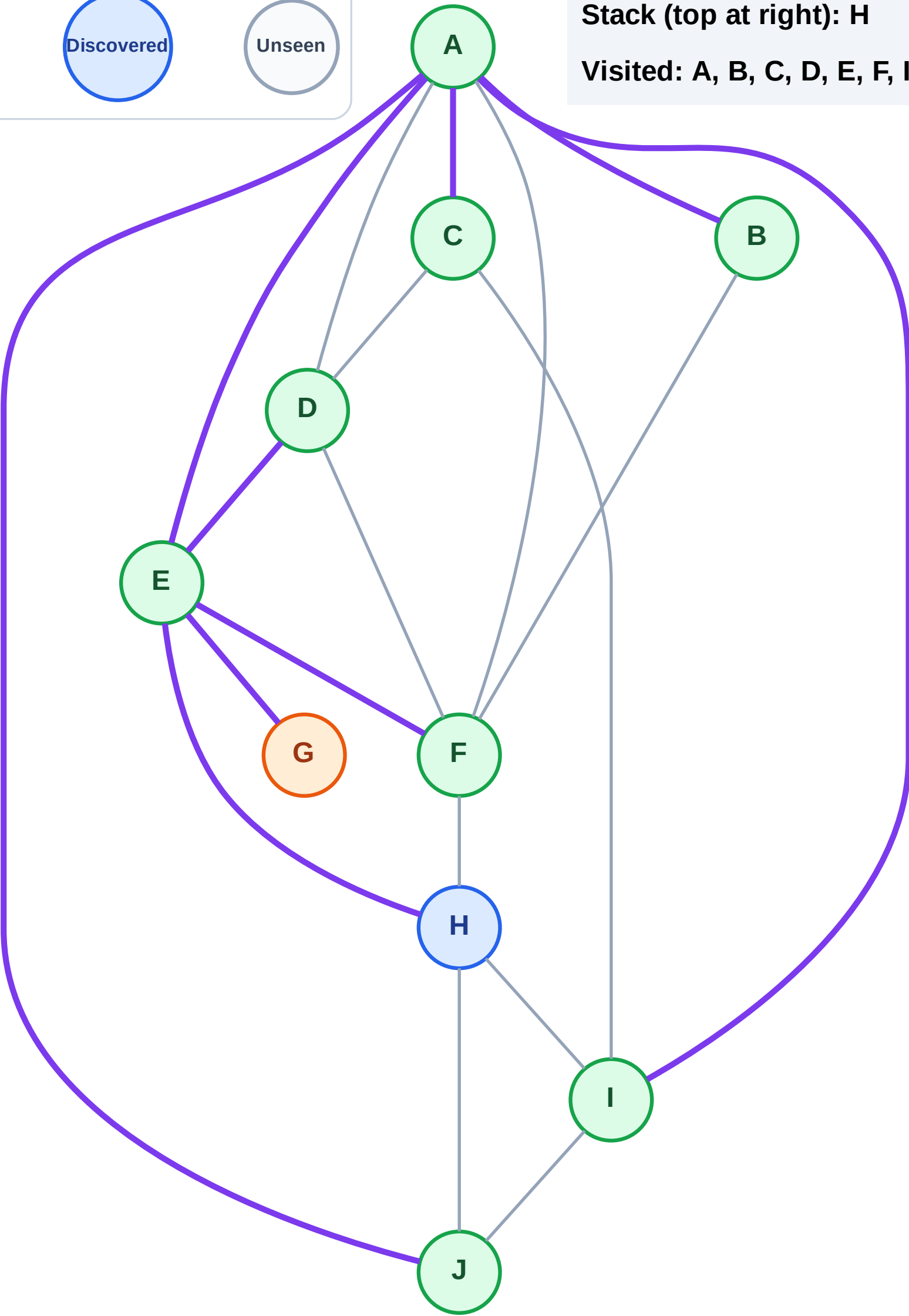
Complete

Processing

Discovered

Unseen

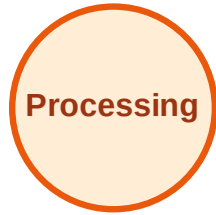
Stack (top at right): H
Visited: A, B, C, D, E, F, I, J



DFS Traversal

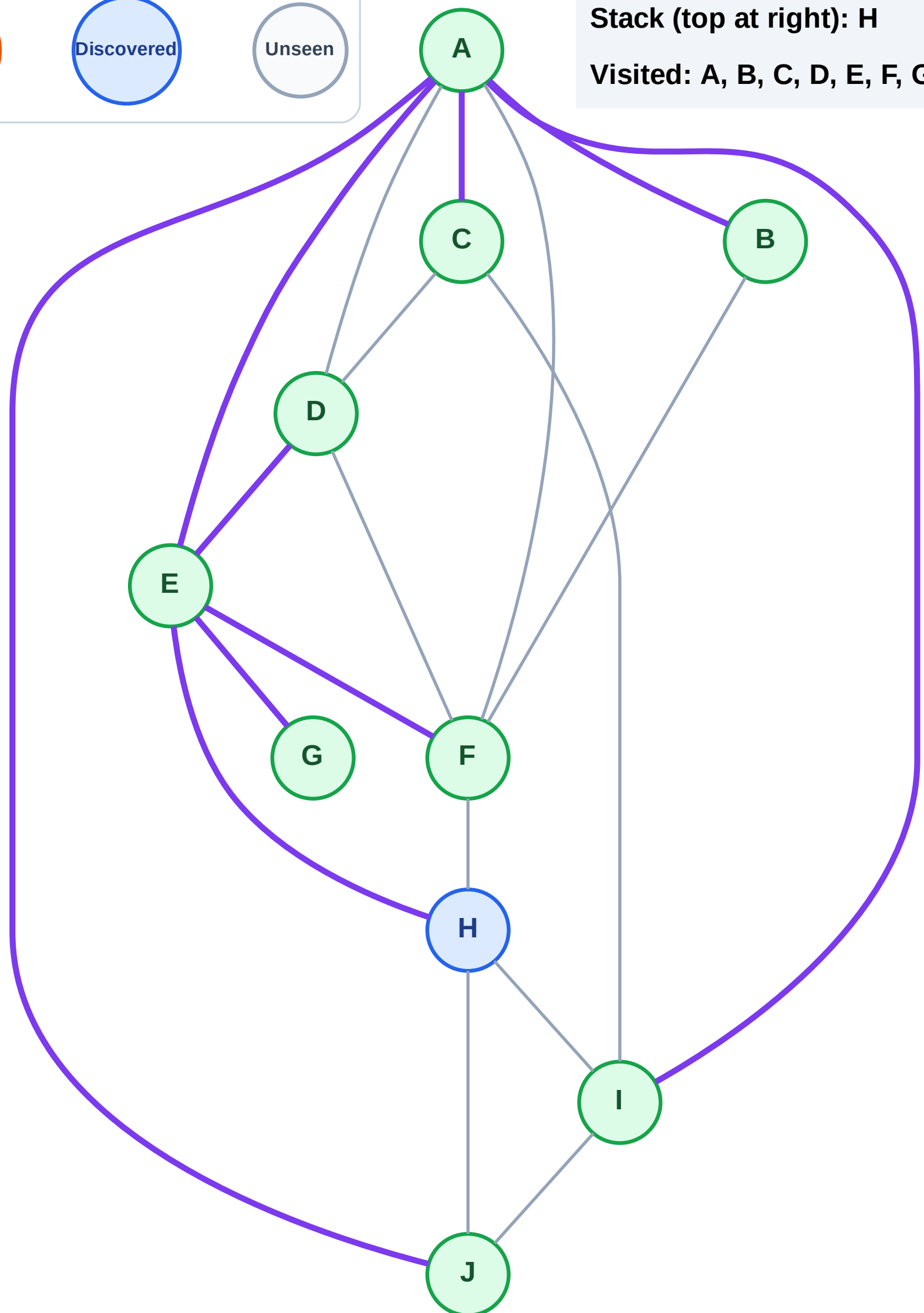
Step 19: No new neighbors from G

Legend



Stack (top at right): H

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

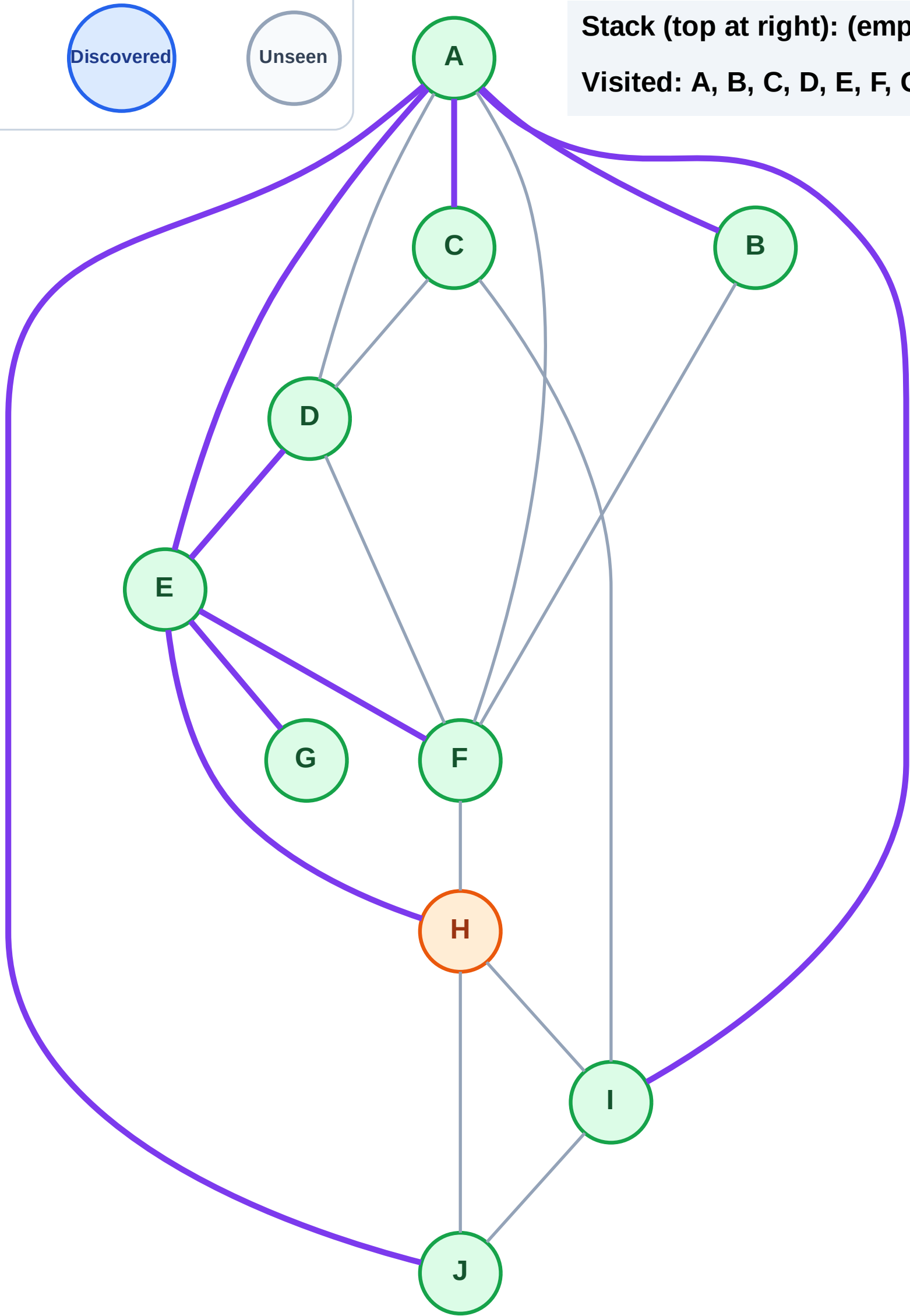
Step 20: Process node H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

